

PennStateSoft Meeting Scheduling System
Project Analysis and Development Plan
Version <1.0>

TEAM MEMBERS

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Revision History

Date	Version	Description	Author
<07/05/26>	<1.0>	Outlined overview aspects of project, created use cases and their respective diagrams, included vulnerability concerns.	Samuel Shumake, Victoria Worthington, Lianjie Sun, Tiffany Hart

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Project Analysis and Development Plan

1. Introduction

The Software Requirements and Specification (SRS) document details the high-level needs and objectives of the PennStateSoft Meeting Scheduling System (PMSS). It defines a set of stakeholders and derives functional and nonfunctional requirements from their needs and expectations. This document will explain why these needs exist and how the derived requirements satisfy them.

1.1 Purpose

The purpose of this document is to define and document the expectations and requirements of PMSS. This will be accomplished by analyzing and ranking stakeholder demands by relevance, then converting them into actionable functional and nonfunctional requirements. These findings will serve as the foundation for the planning, design, and development of the system.

1.2 Scope

This document will be responsible for the defining the primary stakeholders, analyzing their needs and expectations, creating the functional and nonfunctional requirements of the system, and establishing any foreseen standards and procedures needed for the successful planning, design, and development of the project. Additionally, textual and visual models, including use case diagrams, sequence diagrams, and class diagrams, will be provided to support these ends.

1.3 Definitions, Acronyms, and Abbreviations

Acronym	Meaning
SRS	Software Requirements and Specification
PMSS	Project Meeting Scheduling System
SD	Sequence diagram
ADA	Americans with Disabilities Act
HTTPS	Hypertext Transfer Protocol Secure
TLS	Transport Layer Security
ISO	International Organization for Standardization
IEEE	Institute of Electrical and Electronics Engineers

1.4 References

The following documents, standards, and external resources were utilized in the creation of this SRS:

1. U.S. Department of Justice Civil Rights Division. (2022, March 18). Guidance on Web Accessibility and the ADA. ADA.Gov. <https://www.ada.gov/resources/web-guidance/>
2. ISO/IEC/IEEE. (2019, July). 15289:2019: Systems and software engineering — Content of life-cycle information items (documentation). ISO. <https://www.iso.org/standard/74909.html>

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1.5 Overview

The SRS is organized into two primary sections. Section 1 provides a high-level introduction to the PMSS, including the system's purpose, scope, definitions, references, and overall structure of the document. Section 2 contains the problem description and detailed system requirements. This includes the system purpose, functional and nonfunctional requirements, user descriptions, constraints, and specific requirements, such as applicable standards, system requirements, performance requirements, environmental requirements, and quality requirements. This section includes use cases with corresponding extended use case and sequence diagrams, with vulnerability mapping and supporting documentation. A supplementary section is included to provide a team member log sheet, which records individual team member contributions to the SRS.

2. Problem description

2.1 Project Purpose, Scope, and Objectives

The purpose of the Meeting Scheduling System (MSS) is to provide a platform for PennStateSoft to maintain the meeting schedule. The system is designed to replace manual or fragmented scheduling processes and ensure efficient management of meetings within a specific period.

The MSS is a web-based application that supports meeting creation, room reservation, and participant management within a fixed weekly schedule. The system operates only within standard business hours (9 AM to 5 PM) and is limited to managing schedules for a single week at a time. Furthermore, the system does not automatically perform scheduling optimization. Instead, it allows users to manually create and manage meetings while enforcing constraints such as time conflicts, room availability, and access control.

The main objectives of the MSS include:

- Providing efficient meeting management.
- Ensuring proper room allocation and preventing scheduling conflicts.
- Supporting role-based access control for clients and administrators.
- Enhancing system usability so that clients require no training and administrators require minimal training.
- Ensuring system reliability by preventing double-booking of rooms and overlapping meetings for users.
- Maintaining security by enforcing access control policies and protecting sensitive user and billing information.

2.1.1 User interfaces

The MSS will be implemented as a web-based application accessible through various web browsers. Therefore, the user interface will be designed to be intuitive and user-friendly, which means requiring no formal training for clients and minimal training for administrators.

In addition, the system will include the PennStateSoft company logo on all main interfaces to maintain branding consistency, and context-sensitive guidance messages will be provided to assist users during system operation. Furthermore, the interface will support two user roles (clients and administrators), with role-specific dashboards and functionalities. Clients will have access to meeting creation and management tools, while administrators will have access to system configuration, reporting, and complaint handling features.

2.2 Product Functions

2.2.1 Client Functions

2.2.1.1 Register Client Account

The client shall be able to create an account using their unique company email address.

2.2.1.2 Profiles

The client shall be able to view and edit their profile information.

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2.2.1.3 Meeting Creations

The client shall be able to create a meeting and reserve a room for it.

2.2.1.4 Pay for Special Room Reservation

The client shall be able to pay money to reserve a special room.

2.2.1.5 Add Participant to Meeting

The client shall be able to add other clients to meetings they have created.

2.2.1.6 Remove Participant from Meeting

The client shall be able to remove other clients from a meeting they have created.

2.2.1.7 Viewing Meetings

The client shall be able to view and edit all of the meetings they have created.

2.2.1.8 Display Meetings

The client shall be able to display all of the meetings they are going to attend.

2.2.1.9 Filing Complaints

The client shall be able to file a complaint.

2.2.2 Administrator Functions

2.2.2.1 Adding Rooms

The administrator shall be able to add rooms to be available for meetings.

2.2.2.2 Deleting Rooms

The administrator shall be able to delete rooms from the pool of available rooms.

2.2.2.3 View and Edit Billing

The administrator shall be able to view and edit the billing information of clients.

2.2.2.4 View and Respond to Complaints

The administrator shall be able to view and respond to complaints.

2.2.2.5 Display All Meetings

The administrator shall be able to display all meetings in weekly and daily timeframes.

2.2.2.6 Display Specific Rooms

The administrator shall be able to display all meetings in a specific room.

2.2.2.7 Display Client's Meetings

The administrator shall be able to display all meetings attended by a specific person.

2.2.2.8 Display By Time Slot

The administrator shall be able to display all meetings at a specific time slot.

2.2.2.9 Create Administrator Account

The administrator shall be able to create new administrator accounts

2.2.2.10 Generate Reports

The administrator shall be able to generate reports from the database.

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2.3 User Description

2.3.1 User Environment

The application is a Meeting Scheduling System (MSS) designed for use by over 100 employees of PennStateSoft, a game development company located in Pennsylvania. The main office is in Pittsburgh, PA. The application will be usable asynchronously from standard web browsers. No dedicated desktop or mobile application is required. All data will be stored in a centralized database managed by the database administrator. Employees will have access to past meetings they've attended. Meeting coordinators will be able to manage meetings. Administrators will have full access to all system functions and data.

2.3.2 User Profiles

All users are required to be a current PennStateSoft employee to access the system and have basic knowledge of using a web browser. Hints and help information will be available to all users when necessary. There will be over 100 clients using the system asynchronously.

2.3.2.1 Employees

Employees are the target market for the MSS. The employees will not require prior knowledge of the system prior to use. Employees must have employee accounts to access the system. Each employee will have their own account and profile for the system.

2.3.2.2 Meeting Coordinators

Meeting Coordinators are responsible for creating a meeting and reserving rooms prior to said meeting. They oversee payments of any special rooms reserved for a meeting. The Meeting Coordinators will be able to file complaints and add and remove attendees for the meetings they have created.

2.3.2.3 System Administrators

The System Administrators are responsible for managing all meetings, rooms, billing, and complaints. The System Administrators are also responsible for implementing and securing the meeting database for the MSS. They will have the ability to add or remove rooms and are responsible for resolving any ongoing complaints submitted by the Employee or Meeting Coordinator. The System Administrators will require 2 hours of training prior to accessing their functions within the system.

2.4 Assumptions and Dependencies

The following assumptions are made for the development of the Meeting Scheduling System (MSS):

- Every user must have a valid and unique PennStateSoft company email address, which serves as the user's primary identifier.
- Only authorized PennStateSoft employees are permitted to access the system.
- The system manages meeting schedules for one week only, with all meetings restricted to standard business hours (9:00 AM–5:00 PM).
- Meeting schedules are organized into predefined one-hour time slots.
- Each meeting must be assigned to exactly one room.
- Reservations for special rooms require a payment of \$100, and payment is assumed to be successfully processed once confirmation is received through an external or simplified internal payment mechanism.
- The system is accessed through various web browsers.

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- A reliable database is available to permanently store and manage all related items, such as user, meeting, room, and payment information.

The Meeting Scheduling System (MSS) depends on the following external systems and conditions:

- A web server infrastructure is required to host the application and handle concurrent user requests.
- A relational database management system (RDBMS) is required to store and manage persistent data, including user accounts, meetings, rooms, and billing information.
- The system should be accessible through various web browsers.
- An external or internal payment service is required to process payments for special room reservations.
- An email or notification service may be used for account verification and communication with users.
- Accurate organizational data provided by PennStateSoft, including valid employee records and room information maintained by administrators, is required for the system to operate correctly.

2.5 Constraints

The system must be accessible via web browser and user-friendly. Data must be stored on a central database and allow for generation of reports. The application only maintains a single calendar week, and meetings can only occur during business hours (9am – 5pm). The application must be able to support simultaneous access from over 100 people.

2.6 Specific Requirements

2.6.1 Applicable Standards

The system shall comply with applicable local and federal regulations, and relevant industry, communication, web, and quality standards to ensure secure and reliable operation, which include the following:

- The web user interface shall adhere to ADA web accessibility guidelines to improve accessibility and usability for all users.
- The system shall use secure communication through HTTPS, enforcing TLS for all communications between the web browser and application.
- Data storage and handling shall align with corporate data privacy standards, using industry standard encryption methods for sensitive user data, such as passwords and billing data.
- The software development process shall be guided by ISO software quality principles and IEEE recommended practices for fulfilling software requirement specifications.

2.6.2 System Requirements

The PMSS shall be fully functional on the following supported web browsers: Chrome, Firefox, Microsoft Edge, and Safari. The PMSS shall require no additional software installation beyond a supported web browser. The client shall have a minimum of 1 GB RAM and a stable internet connection. The PMSS shall support the following host operating system: Windows 10 or later, MacOS, and Linux.

2.6.3 Performance Requirements

The PMSS shall load pages within 2 seconds with normal network conditions. User actions, such as meeting creation and adding meeting participants, shall be completed within 3 seconds. The PMSS shall support at least 400 simultaneous users without increasing system response times by more than 15% compared to normal operations. The PMSS shall remain 99% available during business hours, with the exception of quarterly scheduled maintenances.

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2.6.4 Environmental Requirements

The PMSS shall be accessible from any supported browser with a stable internet connection, displaying a connectivity message if internet becomes unavailable or unstable. Invalid user inputs shall produce a clear, user-friendly error message and provide guidance for resolving the issue. System/server issues shall display a generic error message. Database backups shall be performed daily to prevent data loss in the event of system failure or corruption. Scheduled maintenance shall occur outside normal business hours quarterly.

2.6.5 Quality Requirements

2.6.5.1 Correctness

The PMSS shall correctly implement all functional requirements described in the SRS. Meeting scheduling guidelines, room availability, user permissions, and payment requirements shall operate without defects in the system's specification, design, or implementation.

2.6.5.2 Reliability

The PMSS shall maintain a maximum failure rate of no more than one unscheduled outage per quarter during standard business hours.

2.6.5.3 Efficiency

The PMSS shall efficiently utilize processing, storage, and network resources to support at least 400 simultaneous users, maintaining page load times of 2 seconds or less and complete user actions within 3 seconds during normal operating conditions.

2.6.5.4 Integrity

The PMSS shall prevent unauthorized access through user authentication and role-specified access control (client vs. admin account). Clients shall only view meetings in which they participate or create, and meeting creators shall only be permitted to modify their created meeting, including adding or removing participants. In addition to client permissions, administrators shall have access to the system to perform administrative functions. User credentials and billing data shall be securely protected during transmission and storage. User sessions shall automatically expire after 30 minutes of inactivity to reduce the risk of unauthorized access.

2.6.5.5 Usability

The PMSS shall require no formal training for clients and no more than 2 hours of training for administrators. The user interface shall provide clear navigation and validation messages.

2.6.5.6 Maintainability

The PMSS shall be designed using modular, well-documented code with low coupling to minimize the time and effort needed to identify, modify, and verify software defects.

2.6.5.7 Flexibility

The PMSS architecture shall be designed to support adaptability, allowing for new features and system capabilities to be added with minimal modification to the existing system.

2.6.5.8 Testability

The PMSS shall maintain audit and error logs for login attempts, meeting modifications, payment transactions, and system errors for software testing and debugging purposes.

2.6.5.9 Reusability

The PMSS shall be developed using modular, object-oriented components that can be reused in future PennStateSoft software projects.

2.6.5.10 Survivability

The minimum time between unscheduled PMSS failures shall be 91 days (an annual quarter), and the system shall be restored to normal operation within 1 hour of the failure. Regular database backups shall ensure that meeting, user account, and billing information are protected from being lost and can be recovered if needed.

2.7 Analysis model

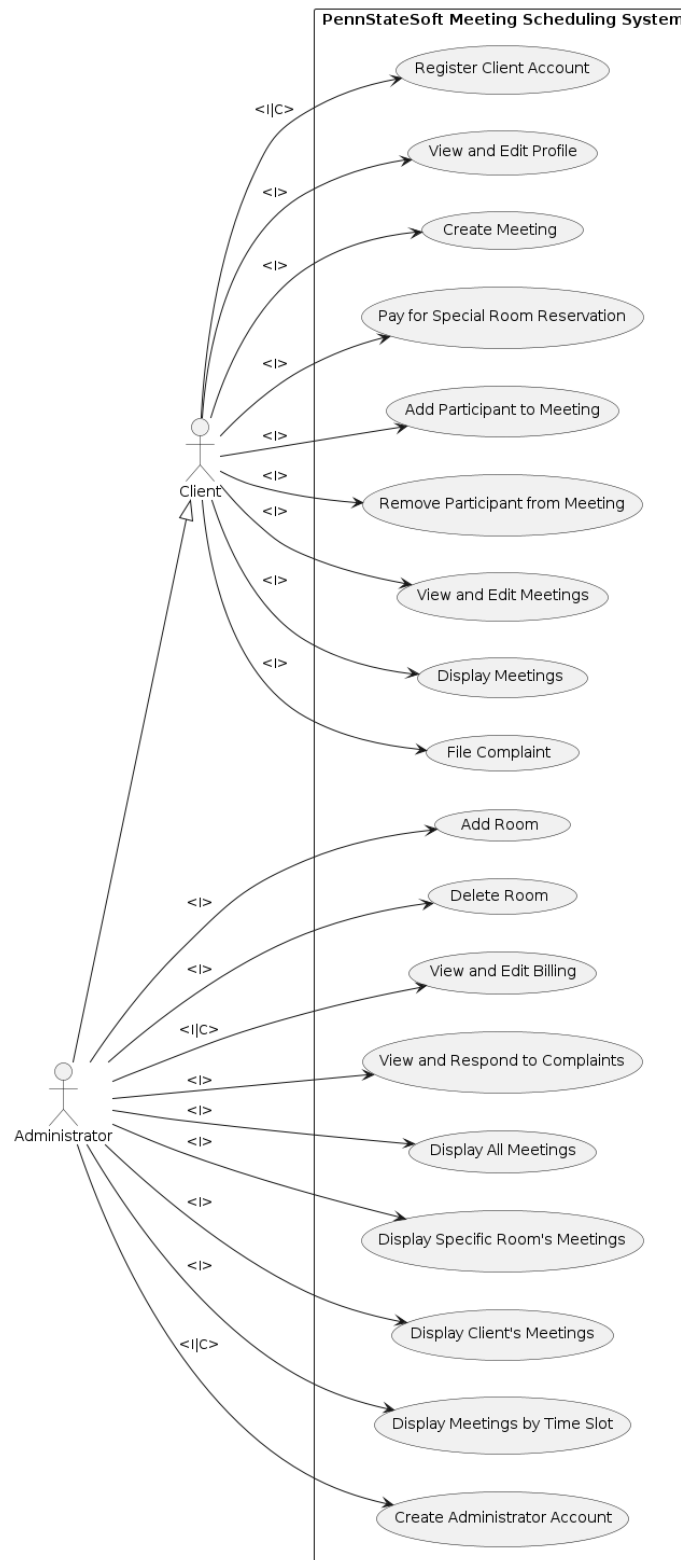


Figure 0: Overview Use Case Diagram

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Use Case ID	Usecase-01
Use Case Name	Register Client Account
User Goal	Allow user to register for client account using their email.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Client
Precondition	Client does not have an active account.
Minimal Guarantee	System does not cause data corruption, returns client to register account page.
Success Guarantee	Client creates active account.
Fail Case	Account not created.
Consequence of Failure	Prospective client is unable to create an account.
Associated Risks	Attack on users database to leak private information such as usernames and passwords.
Trigger	Client clicks the button to create an account.
Steps	
1	System displays account registration form.
2	Client enters email address into email field.
3	Client enters password into password field.
4	Client submits account registration form.
5	System verifies no account with given email exists.
6	System creates the new account.
7	System displays success message to client and redirects them to their main dashboard.
Extension Steps	
2A1	Client enters invalid email address.
2A2	System displays validation error message and prevents form submission.
3A1	Client enters invalid password (e.g. too few characters)
3A2	System displays validation error message and prevents form submission.
5A1	An account with the submitted email already exists.
5A2	System prints an error message informing user an account has already registered that email address.
5A3	System cancels execution of account creation.
6A1	System cannot successfully create account.
6A2	System prints a message asking client to contact an administrator and try again later.

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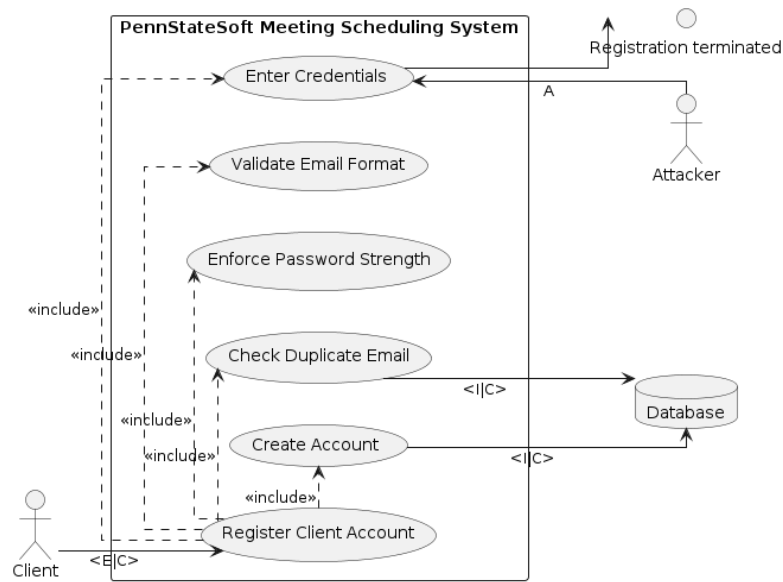


Figure 1.1: Register Client Account Use Case Diagram

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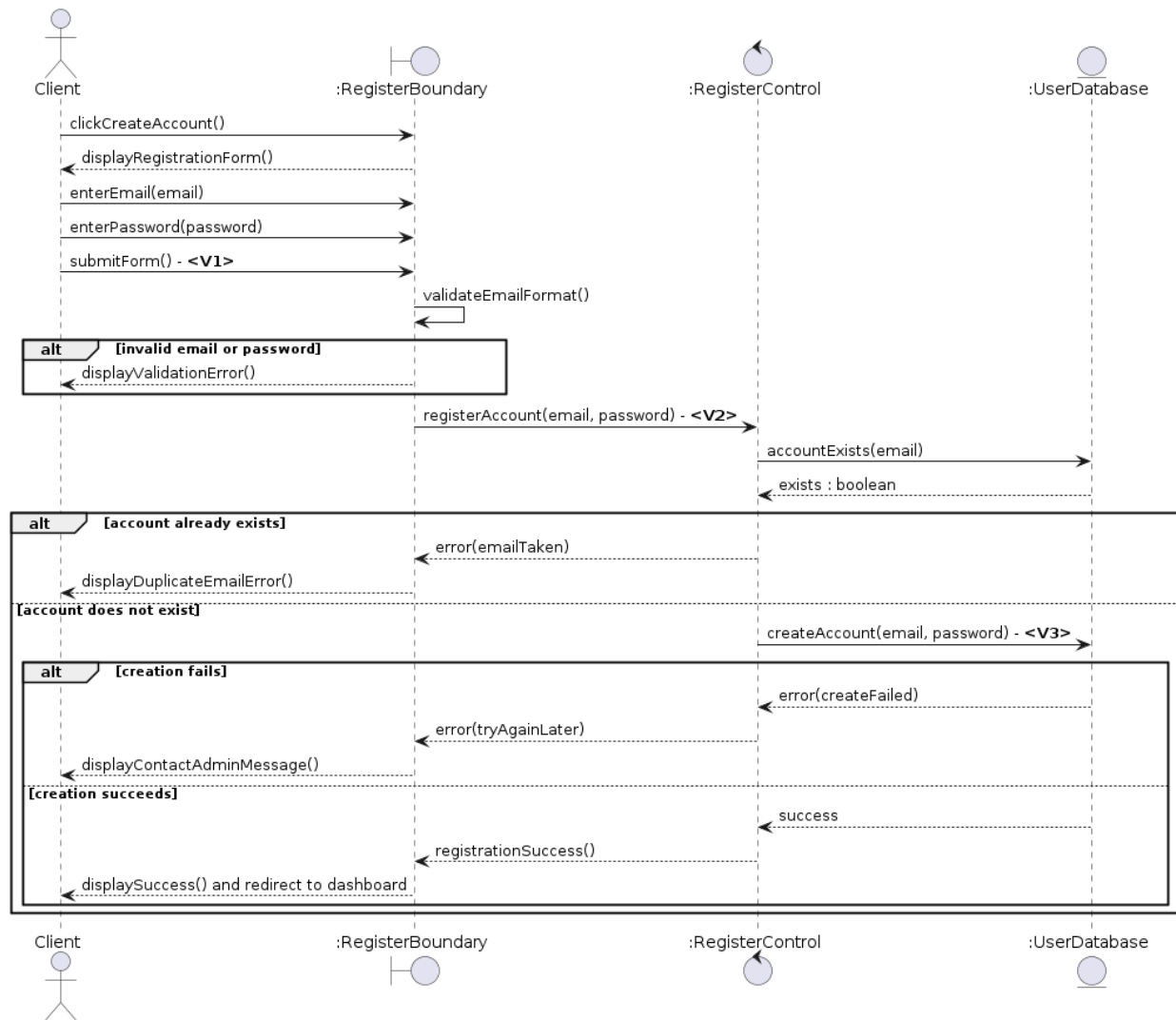


Figure 1.2: Register Client Account Sequence Diagram

Use Case ID	Usecase-02
Use Case Name	Update Profile
User Goal	Allow clients to view and edit their personal information within their profile.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Client
Precondition	Client has an active account within the system.
Minimal Guarantee	System does not cause data corruption; the system does not modify the profile information if an error occurs. Default or previously saved profile must remain intact.
Success	Client successfully views and updates their profile information and the information is stored

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Guarantee	securely.
Fail Case	Client is unable to update profile due to inappropriate display name, profile picture, or invalid birth date.
Consequence of Failure	Client is unable to update their profile and will receive the appropriate error message.
Associated Risks	Unauthorized access to client data; Exposure of client information
Trigger	Client clicks the “view profile” button, makes changes, and clicks “save” to save updated profile.
Steps	
1	System displays the home page for the client.
2	Client selects “view profile”
3	The system displays the client’s profile page.
4	The client updates their profile information
5	The system securely stores new information from the client.
Extension Steps	
4A1	The client tries to change their display name to something nonprofessional.
4A2	The system sends an error that the client display name is invalid.
4B1	The client tries to change the profile picture to something that is not related to the client.
4B2	The system flags the image as inappropriate and returns an error message to the client.
4C1	The client tries to change their birth date to a date that is underaged.
4C2	The system sends an error that the date chosen is invalid.

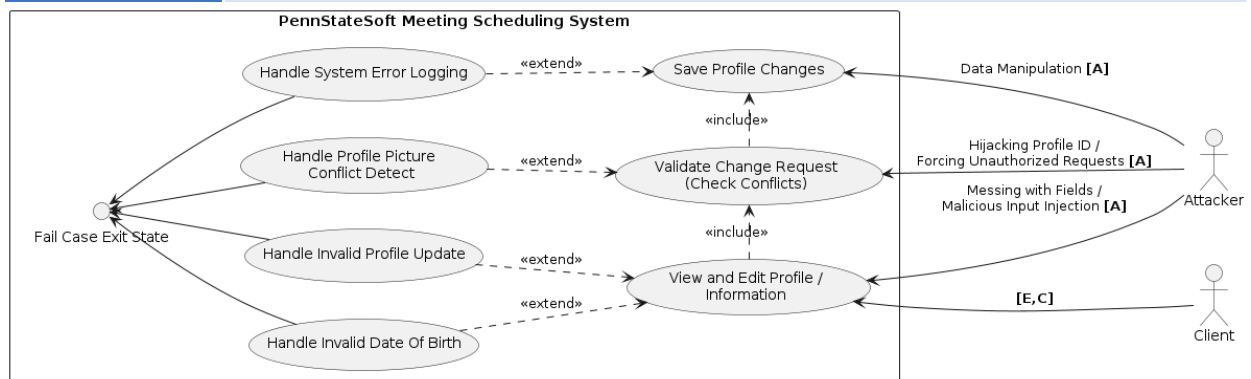


Figure 2.1: Update Profile Use Case Diagram

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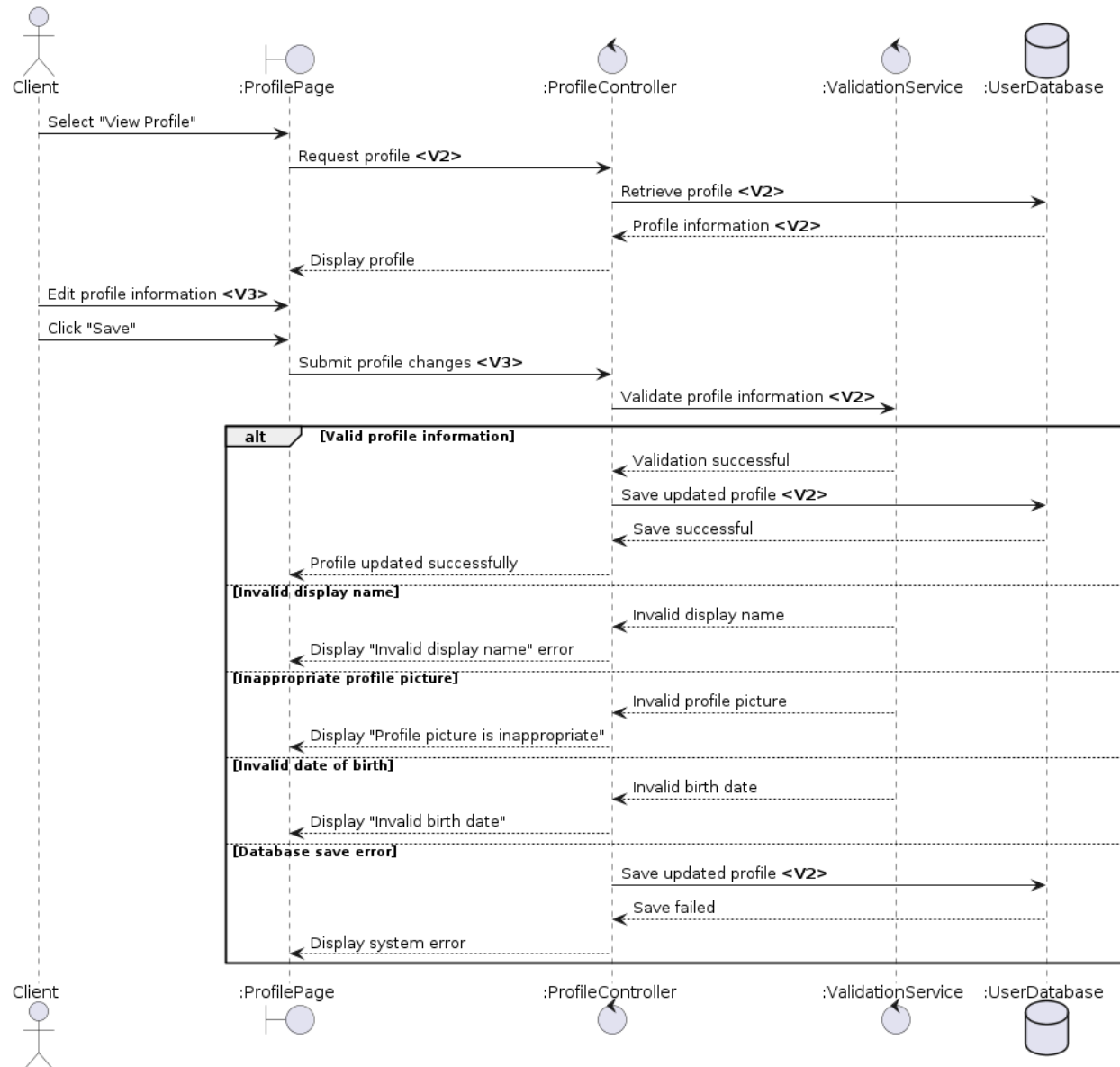


Figure 2.2: Update Profile Sequence Diagram

Use Case ID	Usecase-03
Use Case Name	Create a Meeting and Reserve a Room
User Goal	Client creates a meeting and reserves an available room for the selected time slot.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Client
Precondition	The client is logged in.
Minimal	No invalid meeting is stored in the system; system state remains consistent even if the

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Guarantee	operation fails.
Success Guarantee	A meeting is successfully created, assigned to an available room, and stored in the database.
Fail Case	A user creates a meeting with invalid information, reserves an unavailable room, or attempts to create a meeting that violates scheduling rules.
Consequence of Failure	The room schedule may become inconsistent, a room may be double-booked, or a client may be assigned to multiple meetings at the same time.
Associated Risks	Room double-booking; User schedule conflict; Invalid meeting data; Unauthorized or inconsistent meeting creation
Trigger	The client selects “Create Meeting”.
Steps	
1	User selects Create Meeting.
2	User enters meeting name, date, and one-hour time slot.
3	User selects an available room.
4	System validates business hours (9 AM–5 PM) and input completeness.
5	System checks room availability for the selected time slot.
6	System checks whether the client has conflicting meetings.
7	System creates the meeting and assigns the room.
8	System stores the meeting in the database.
9	System displays confirmation message.
Extension Steps	
4A1	User is not the creator of the meeting; system rejects the request and logs unauthorized attempt.
5A1	One or more selected clients do not exist; system prompts user to select valid client accounts.
6A1	One or more selected clients have scheduling conflicts; system rejects conflicting clients and notifies the user.
7A1	Duplicate participants detected; system removes duplicates automatically or requests user correction.
3A1	No participant is selected; system prompts user to select at least one client before submission.

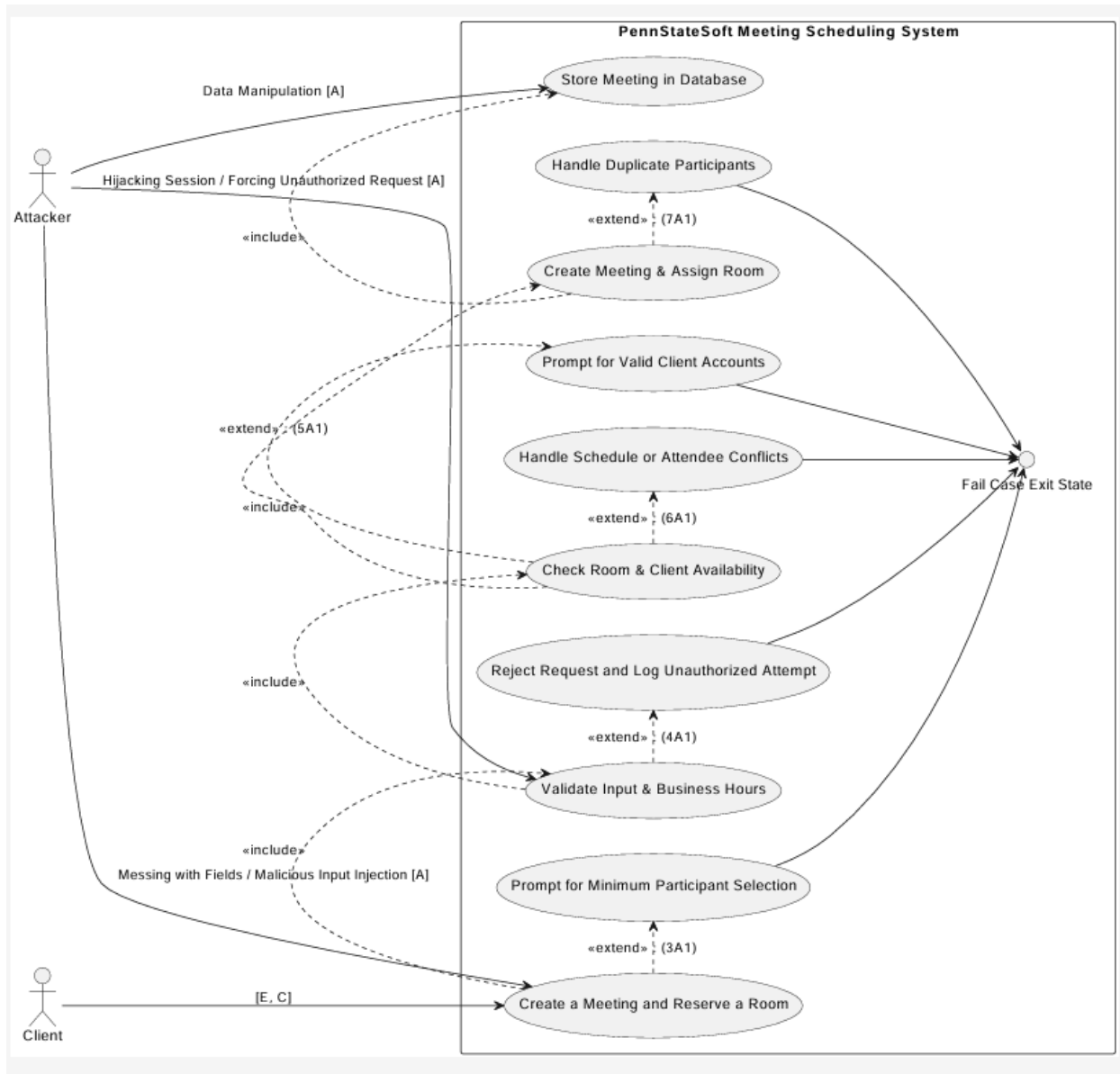


Figure 3.1: Meeting Creation and Reservation Use Case Diagram

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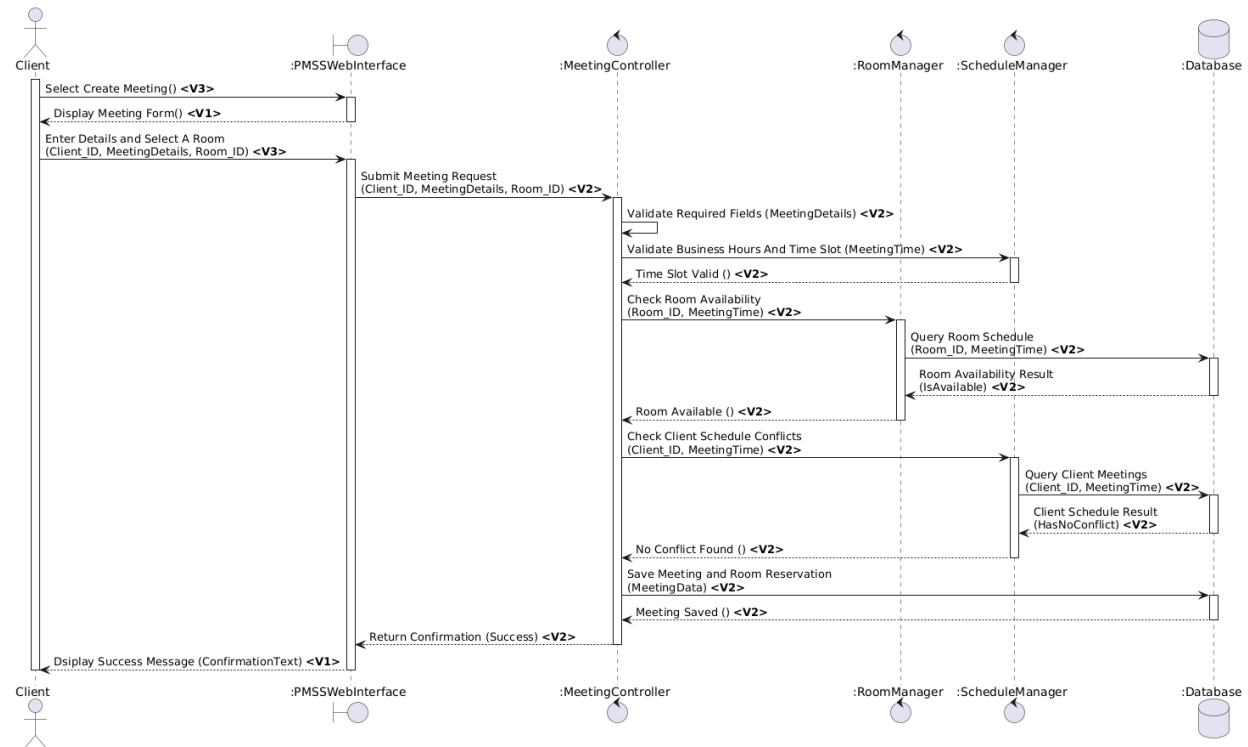


Figure 3.2: Meeting Creation and Reservation Sequence Diagram

Use Case ID	Usecase-04
Use Case Name	Pay for Special Room Reservation
User Goal	Allow client to pay for a special room in order to reserve it for a meeting.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Client
Precondition	Client is authenticated and logged into the system and has created a meeting that requires a special room reservation
Minimal Guarantee	System does not process incorrect charges or duplicate payments and returns client to the payment page if a failure occurs.
Success Guarantee	Client successfully pays the required fee and the special room is reserved for the selected meeting.
Fail Case	Client enters invalid billing data; Payment service denies transaction; Room is unavailable.
Consequence of Failure	The special room reservation is cancelled and the payment is refunded if processed. Room becomes available for other users to select.
Associated Risks	A different client may reserve the exact time and room. Attackers may alter transactions and produce fake error/success signals. Attackers may access billing information.
Trigger	Client selects a special room and clicks the button for a special reservation.
Steps	
1	System displays payment page for special room reservations.

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2	Client selects payment method.
3	Client enters/confirms billing information.
4	Client submits payment request.
5	System validates billing formation.
6	System processes payment through the system payment service.
7	System confirms successful payment.
8	System creates special reservation by reserving a soecial room for the meeting.
9	System displays special reservation confirmation message to the client.
Extension Steps	
3A1	Client enters invalid or incomplete billing information.
3A2	System displays validation error message to client and does not submit payment.
5A1	Billing information fails validation.
5A2	System displays validation error message to client and sends request to update billing information.
6A1	Payment fails due to payment service error.
6A2	System cancels special room reservation request.
6A3	System displays system error to client that payment could not be completed.
8A1	Special room is no longer available.
8A2	System cancels reservation.
8A3	System notifies and provides client with option to select another available room or trigger refund process.

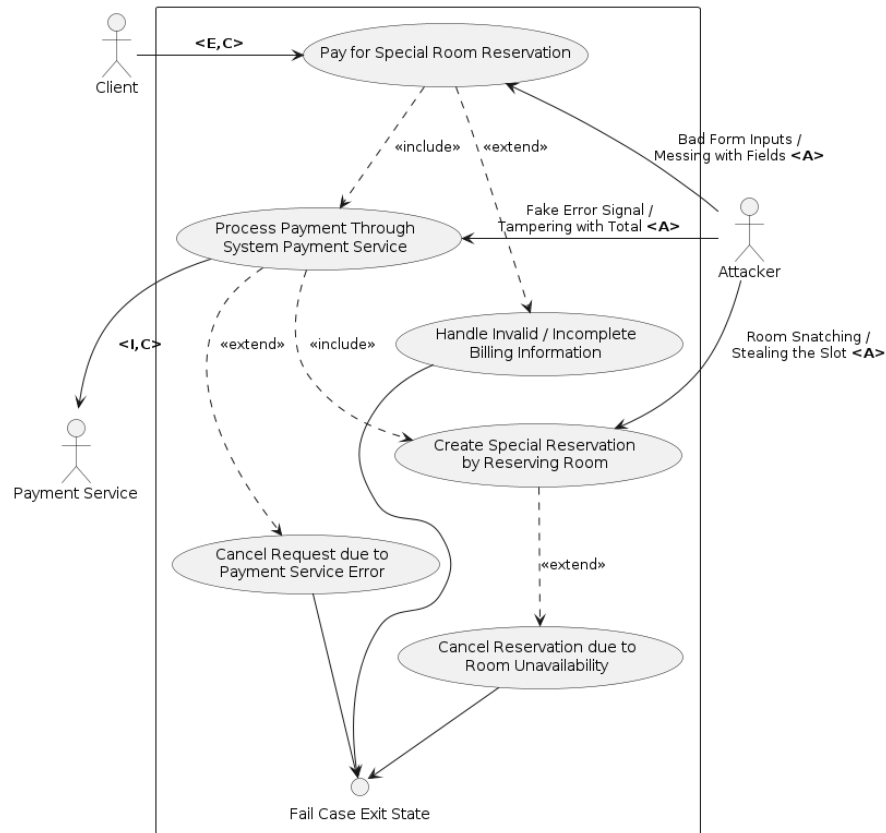


Figure 4.1: Special Meeting Reservation Use Case Diagram

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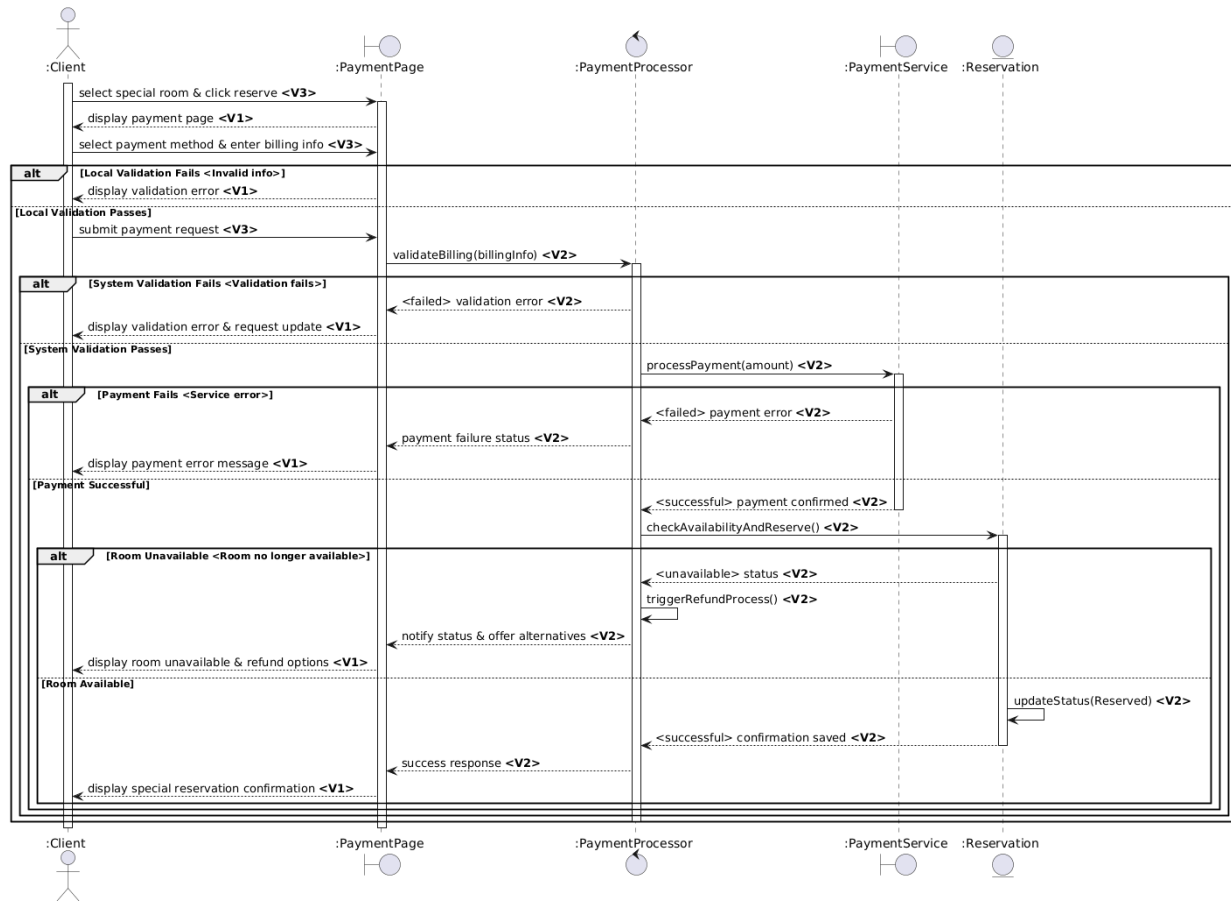


Figure 4.2: Special Meeting Reservation Sequence Diagram

Use Case ID	Usecase-05
Use Case Name	Add Participants to a Created Meeting
User Goal	The client adds other clients to a meeting they have created.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Client
Precondition	The client is logged in and is the creator of the selected meeting.
Minimal Guarantee	No unauthorized modification is made to meeting participant data; system state remains consistent if the operation fails.
Success Guarantee	Selected valid participants are successfully added to the meeting and stored in the database.
Fail Case	A user attempts to add clients to a meeting they did not create, add invalid clients, or add clients who have scheduling conflicts.
Consequence of Failure	Unauthorized users may modify meeting information, invalid participants may be added, or clients may be scheduled for multiple meetings at the same time.

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Associated Risks	Unauthorized modification of meeting participants; User schedule conflict; Invalid or nonexistent participant accounts; Violation of access control rules
Trigger	The client selects “Add Participants” for an existing meeting.
Steps	
1	User selects one of the meetings they have created.
2	User selects Add Participants.
3	User enters or searches for clients to be added.
4	System verifies that the user is the creator of the meeting.
5	System validates that selected clients exist in the system.
6	System checks each selected client for scheduling conflicts.
7	System verifies that no duplicate participants are added.
8	System updates the meeting participant list.
9	System stores the updated meeting information in the database.
10	System displays a confirmation message to the user.
Extension Steps	
1A1	User is not the creator of the meeting; system rejects the request.
3A1	No participant is selected; system prompts user to select at least one valid client.
5A1	One or more selected clients do not exist; system requests the user to choose valid client accounts.
6A1	One or more selected clients have scheduling conflicts; system rejects conflicting clients and notifies the user.
7A1	Duplicate participants detected; system automatically removes duplicates or requests user correction.

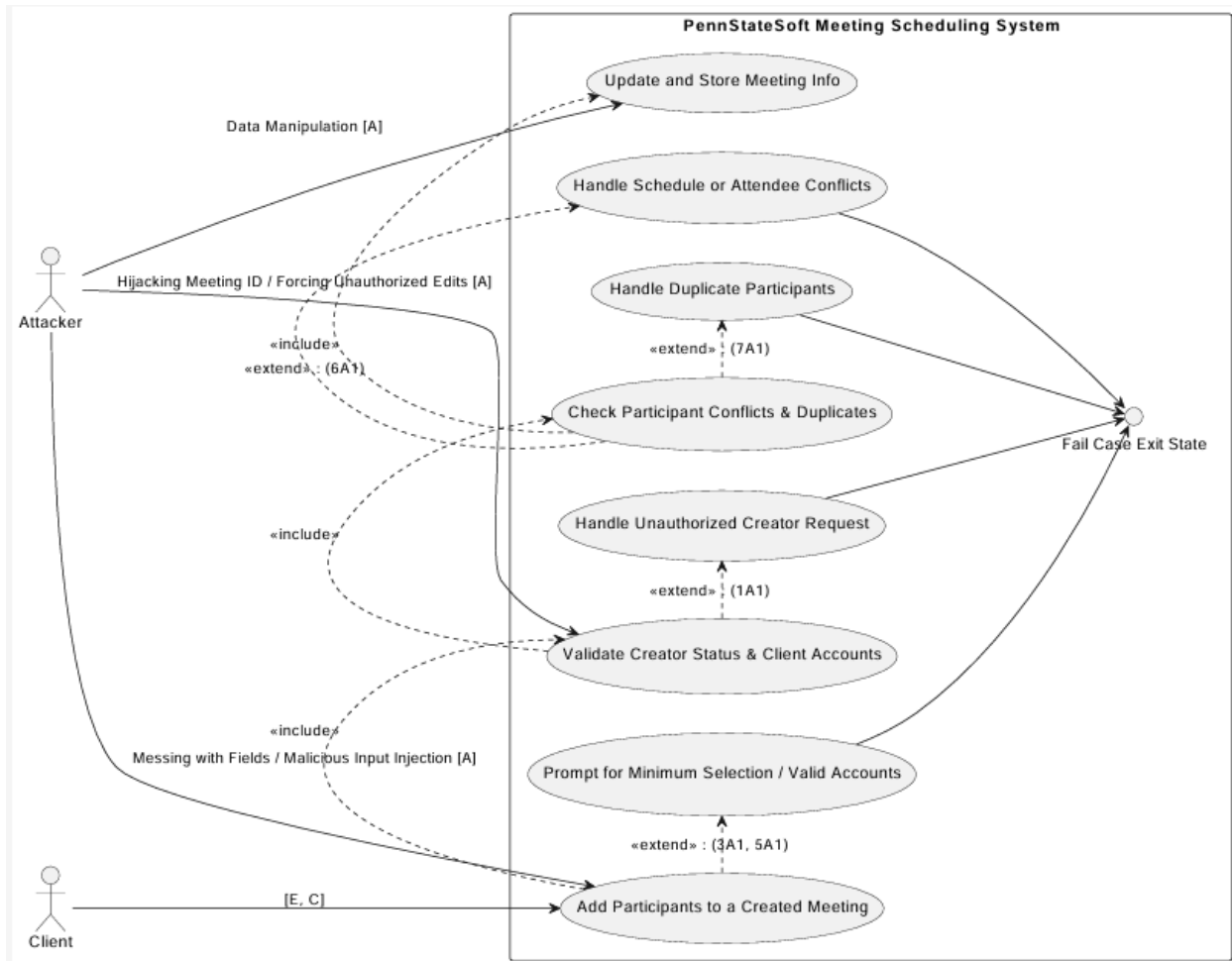


Figure 5.1: Add Participants Use Case Diagram

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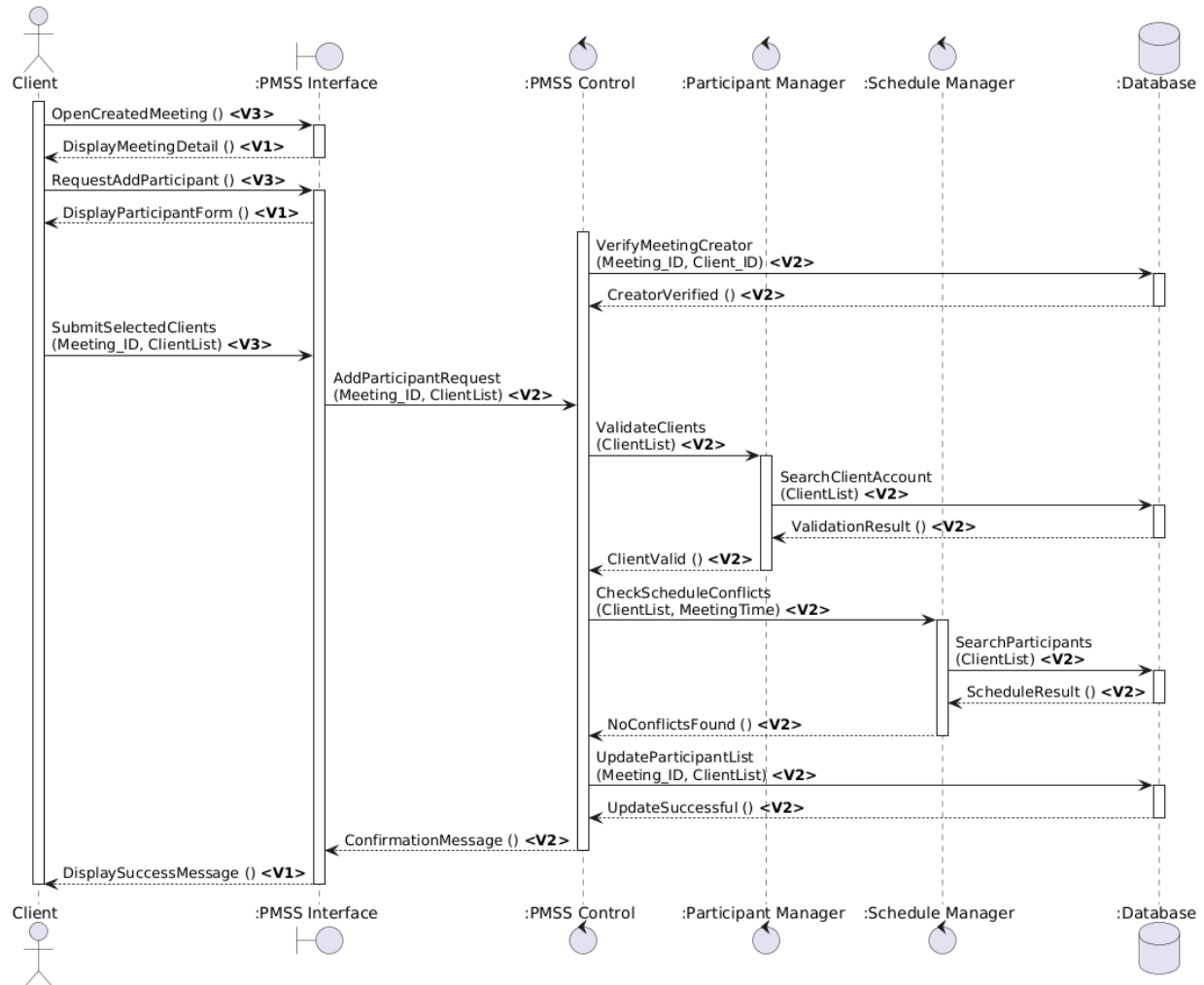


Figure 5.2: Add Participants Sequence Diagram

Use Case ID	Usecase-06
Use Case Name	Remove Participant from Meeting
User Goal	Allow client to remove a participant from a meeting they have created.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Client
Precondition	Client owns a meeting and there is at least one other participant attending.
Minimal Guarantee	System does not cause data corruption, returns client to meeting edit page.
Success Guarantee	Client creates removes specified participant from meeting.
Fail Case	Participant not removed from meeting.

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Consequence of Failure	Unwanted participant still has access to meeting and its details.
Associated Risks	Minimal, unwanted participant can see information on meeting.
Trigger	Client clicks the button to edit meeting participants.
Steps	
1	System displays list of all meeting participants.
2	Client clicks the “Remove” button next to the desired participant.
3	System displays a form asking the user to confirm the removal.
4	Client clicks the “Yes” button to confirm removal.
5	System removes participant from the meeting.
6	System displays a success message informing client of participant removal.
Extension Steps	
2A1	Client tries to remove themselves from a meeting.
2A2	System displays an error message informing them they can’t remove themselves and prevents form submission.
4A1	Client clicks the “No” button to cancel removal
4A2	System cancels form submission and redirects user back to the meeting edit page.

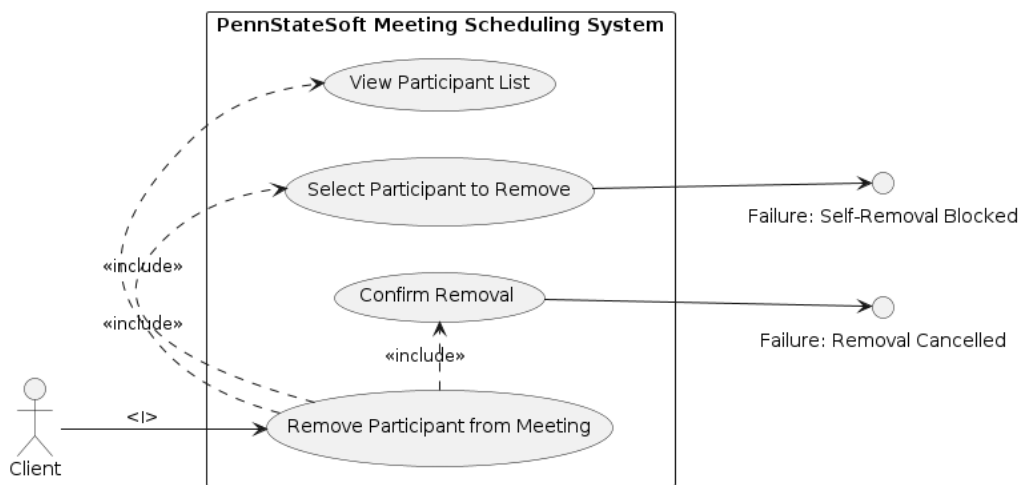


Figure 6.1: Remove Participants Use Case Diagram

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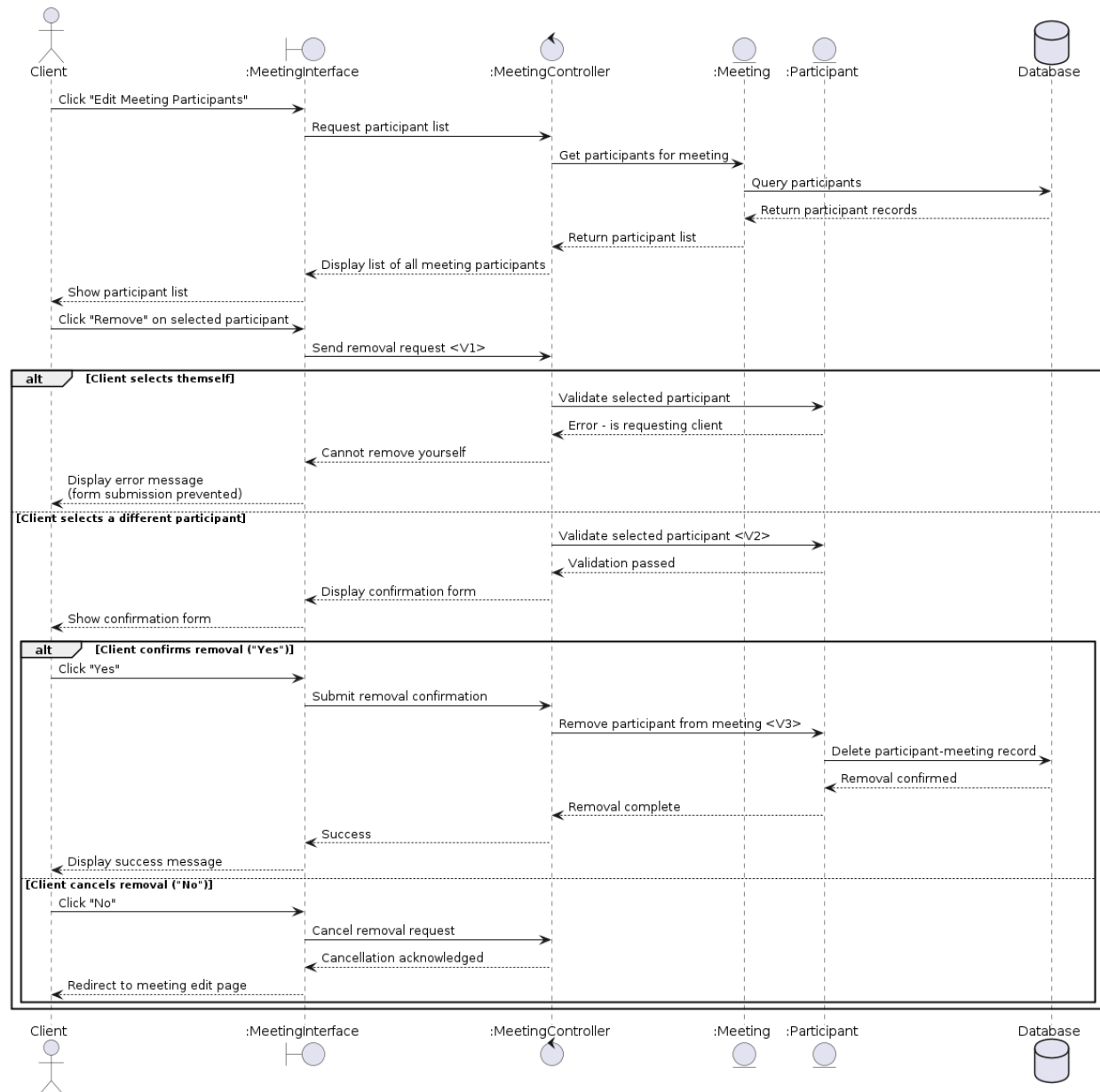


Figure 6.2: Remove Participants Sequence Diagram

Use Case ID	Usecase-07
Use Case Name	View and Edit Created Meetings
User Goal	Allow client to view and edit meetings they have created
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Client
Precondition	Client is logged into the system and has created at least one meeting.

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Minimal Guarantee	System does not modify meeting information if an error occurs or if the client cancels edit. Existing meeting data is intact.
Success Guarantee	Client views and updates the selected meeting and changes are saved.
Fail Case	Client provides invalid data; Scheduling conflict occurs since attendee is already booked for another meeting; System experiences error.
Consequence of Failure	The system rejects modification request and preserves original meeting.
Associated Risks	Attackers may manipulate form inputs to hijack meetings. Meeting data transmission may be interrupted and orphan request.
Trigger	Client clicks the button to edit meetings
Steps	
1	System displays list of all meetings created by the client.
2	Client selects a meeting to view
3	System displays meeting details.
4	Client selects option to edit meeting.
5	Client updates meeting details.
6	Client submits updated meeting information.
7	System validates change request.
8	System saves meeting changes.
9	System displays confirmation message that meeting information is updated.
Extension Steps	
1A1	Client does not have any created meetings.
1A2	System displays invalid message that no meetings are available.
5A1	Client provides invalid/incomplete meeting information.
5A2	System displays a validation error message and does not submit request until information is corrected.
7A1	Updated meeting time conflicts with another meeting scheduled at the selected time and room.
7A1	System displays error message and requests client choose another room/time.
7B1	Attendee/s is already scheduled for another meeting during selected time.
7B2	System displays error message and requests client choose another room/time.
8A1	System has an error while saving updated meeting information.
8A2	System displays error message that client try again later.
8A3	System logs error and does not change meeting information.

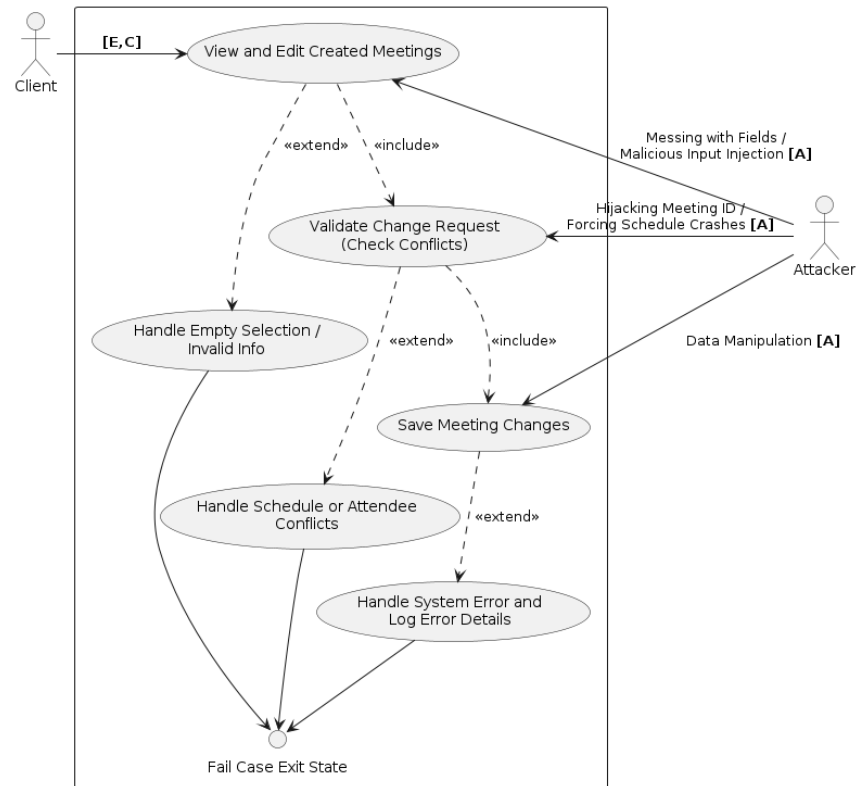


Figure 7.1: Edit Meeting Use Case Diagram

PennStateSoft	Version: 1.0
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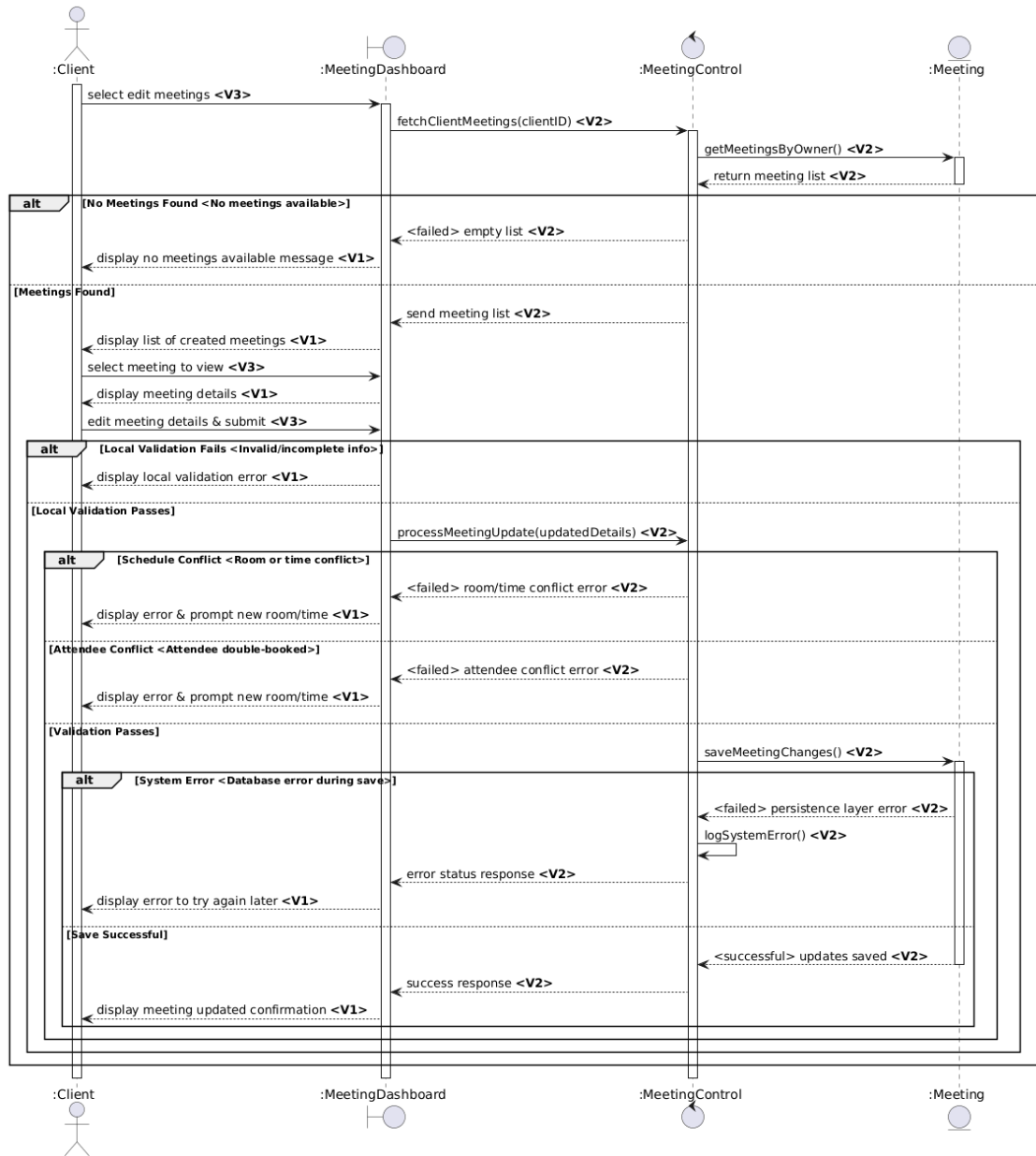


Figure 7.2: Edit Meeting Sequence Diagram

Use Case ID	Usecase-08
Use Case Name	View planned meetings to attend
User Goal	The client views all meetings they are scheduled to attend.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Client

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Precondition	The client is authenticated and logged into the system.
Minimal Guarantee	No unauthorized access to meeting data occurs; system ensures data privacy even if retrieval fails.
Success Guarantee	The system successfully retrieves and displays all meetings the client is scheduled to attend.
Fail Case	System fails to retrieve meeting data due to database error or unauthorized access attempt.
Consequence of Failure	Client is unable to view scheduled meetings or may receive incomplete meeting information.
Associated Risks	Unauthorized access to meeting data; Exposure of other users' meeting schedules; Data inconsistency in meeting retrieval
Trigger	The client selects "View My Meetings".
Steps	
1	User selects View My Meetings.
2	System retrieves the client identity from the active session.
3	System queries the database for meetings where the client is a participant.
4	System filters meetings within the current week schedule.
5	System retrieves meeting details (name, room, time, participants).
6	System displays the list of upcoming meetings to the client.
7	User views meeting details.
Extension Steps	
1A1	Client has no scheduled meetings; system displays an empty list message.
2A1	Session has expired; system redirects user to login page.
3A1	Database query fails; system displays error message and prompts user to retry.

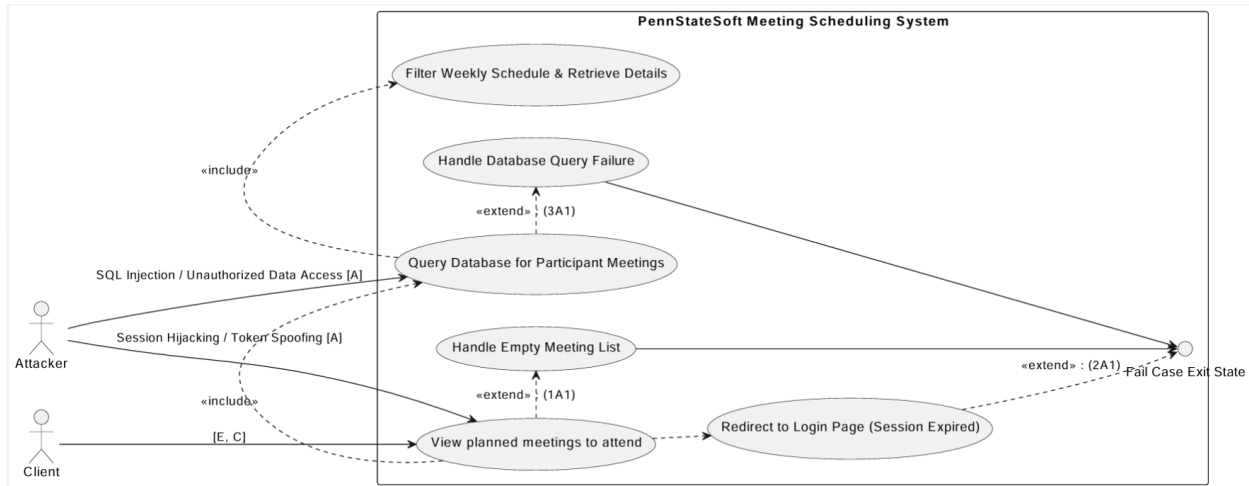


Figure 8.1: Planned Meeting Attendance Use Case Diagram

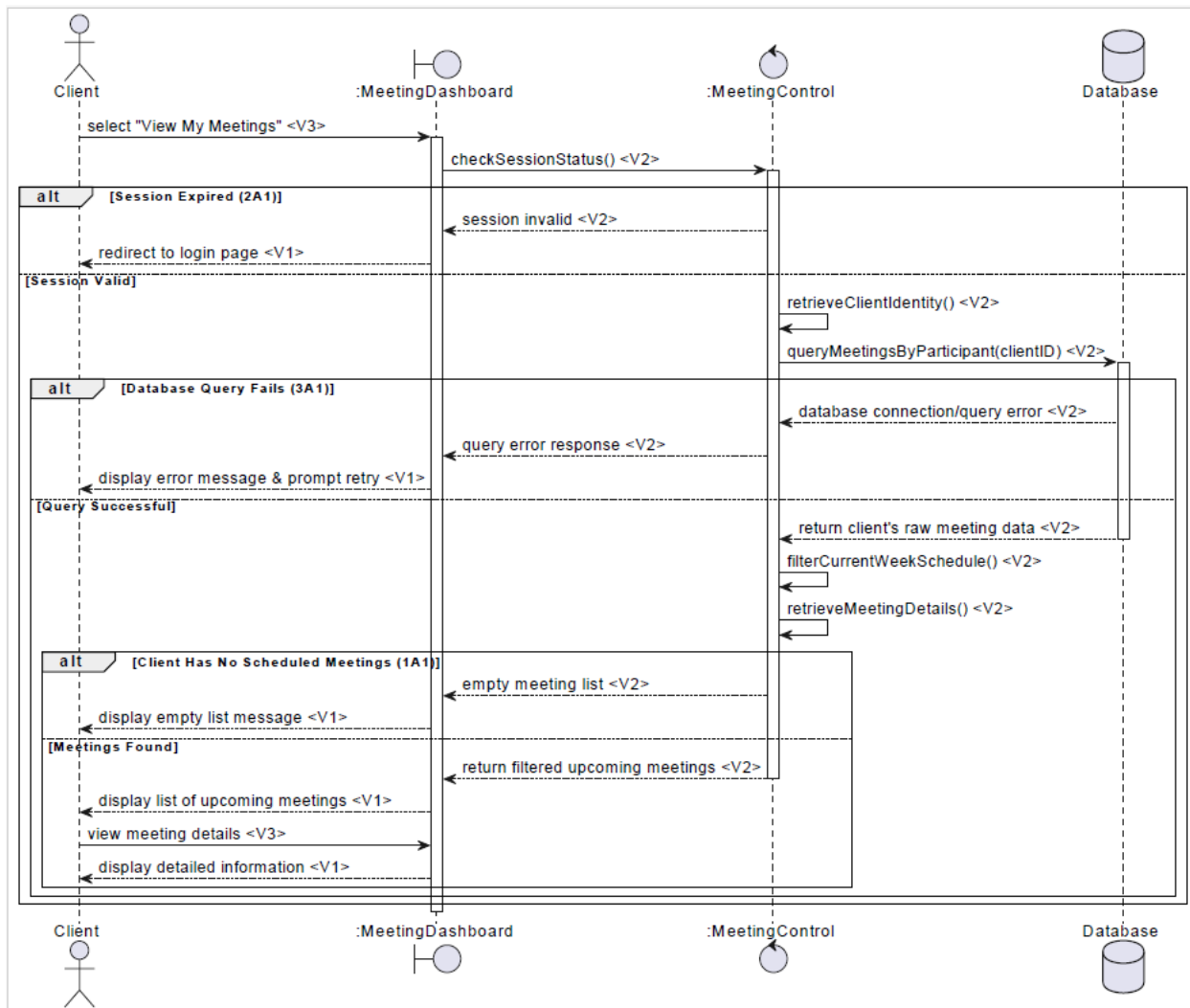


Figure 8.2: Planned Meeting Attendance Sequence Diagram

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Use Case ID	Usecase-09
Use Case Name	File A Complaint.
User Goal	Allow the client to file a complaint.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Client
Precondition	Client has an active account within the system. The client is logged in. The client has attended a minimum of one meeting.
Minimal Guarantee	System does not cause data corruption; returns client to dashboard.
Success Guarantee	Client successfully submits a complaint. The system saves the complaint to the database for the administrator to review.
Fail Case	System fails to save the complaint submitted
Consequence of Failure	Unauthorized access to client data; Exposure of other client's meeting schedules
Associated Risks	Unauthorized access to meeting data; Exposure of client's meeting schedules; Data inconsistency in meeting retrieval
Trigger	Client clicks the "file complaint" button on the dashboard.
Steps	
1	Client clicks the file complaint button.
2	Client chooses the meeting the complaint is regarding.
3	Client chooses from a list of predefined complaint options.
4	Client adds their own information into the summary section.
5	Client presses submit.
6	System saves the complaint to a database for the administrator to review.
Extension Steps	
2A1	Client has not attended any meetings.
2A2	The system will not allow a complaint filed without a meeting being chosen. The submit button remains inactive.
3A1	Client doesn't choose from the predefined complaint options.
3A2	The system will not allow a complaint filed without a complain option chosen. The submit button remains inactive.

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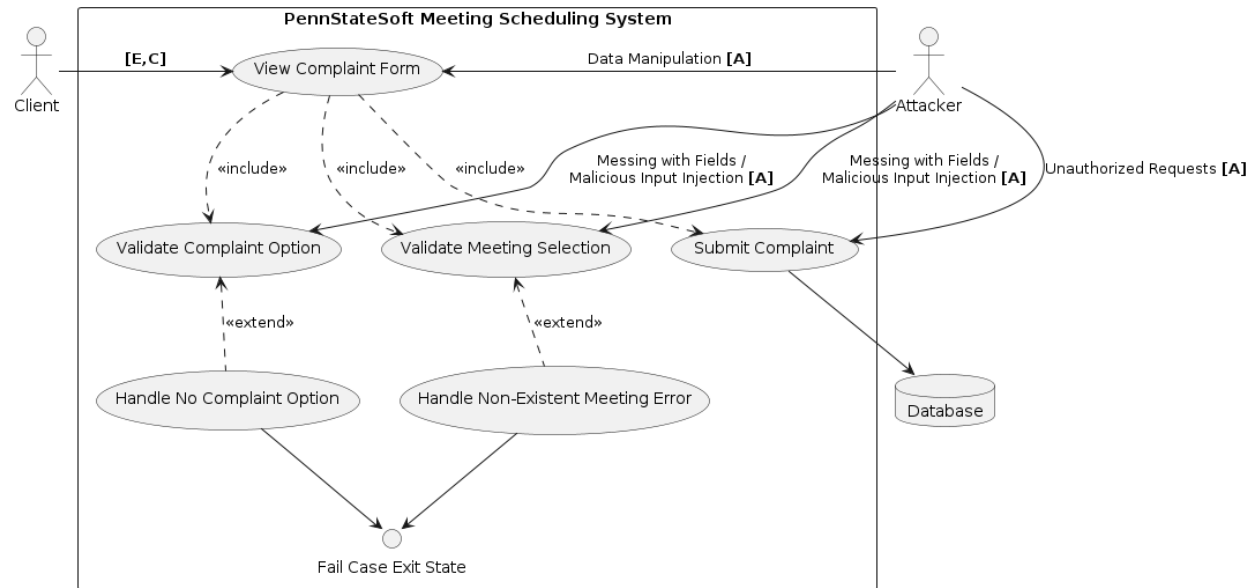


Figure 9.1: File Complaint Use Case Diagram

PennStateSoft	Version: 1.0
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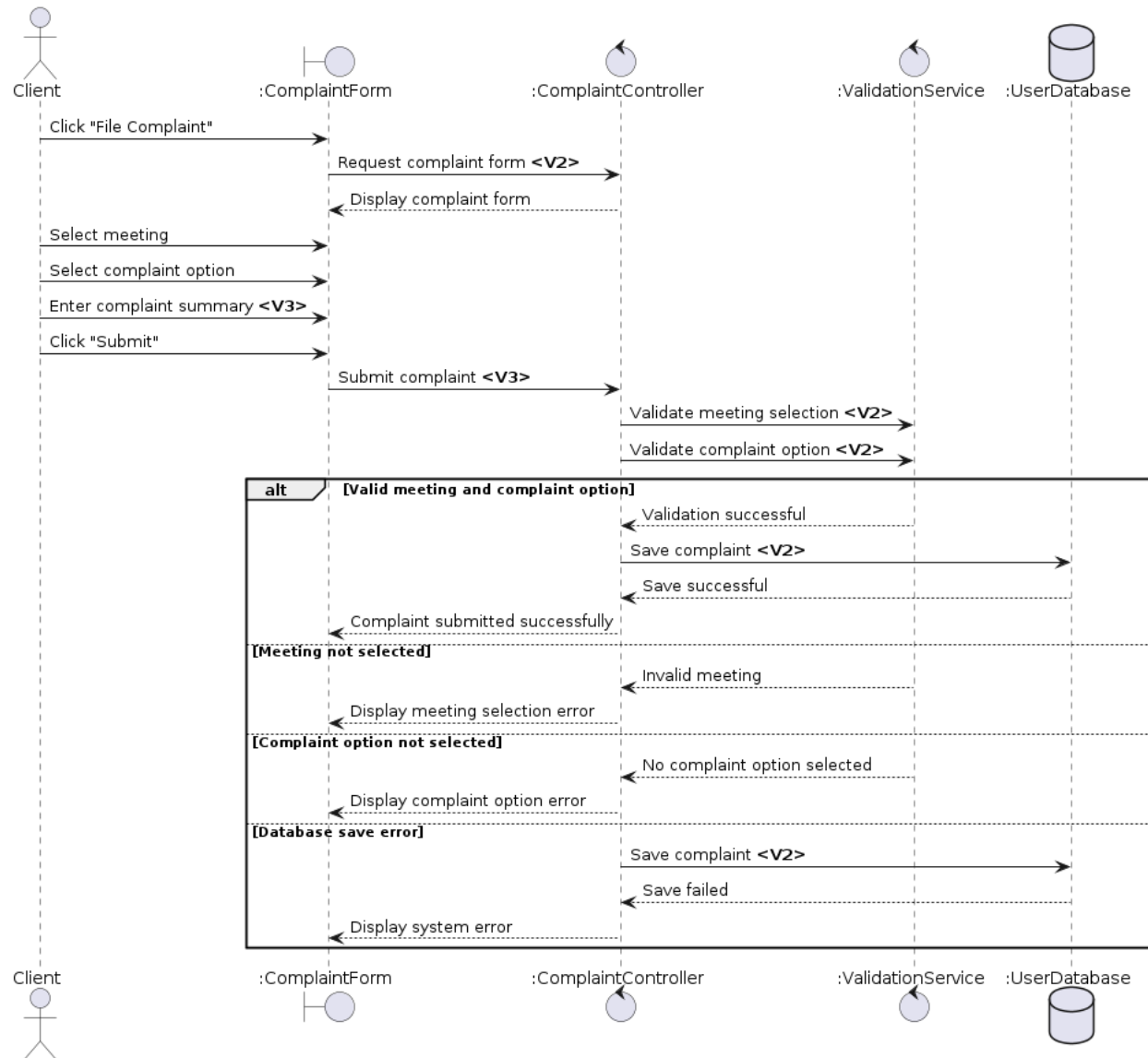


Figure 9.2: File Complaint Sequence Diagram

Use Case ID	Usecase-10
Use Case Name	Add Rooms To the System
User Goal	The administrator adds new rooms to the system for meeting scheduling.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Administrator
Precondition	The administrator is authenticated and has system access privileges.
Minimal	No invalid or duplicate room data is stored; system remains consistent if operation fails.

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Guarantee	
Success Guarantee	A new valid room is successfully added and stored in the database.
Fail Case	An unauthorized user adds a room, or an administrator adds invalid or duplicate room information.
Consequence of Failure	The room list may contain duplicate or invalid rooms, unauthorized users may modify system resources, or meeting reservations may become inconsistent.
Associated Risks	Unauthorized room management; Duplicate room records; Invalid room information; Incorrect room availability for meetings
Trigger	The administrator selects “Add Room” in the room management module.
Steps	
1	Administrator selects Manage Rooms.
2	Administrator selects Add Room.
3	Administrator enters room number and room type.
4	System validates room information (format and completeness).
5	System checks whether the room number already exists.
6	System verifies that the room type is valid (regular or special).
7	System creates a new room entry.
8	System stores the room information in the database.
9	System displays a confirmation message.
Extension Steps	
4A1	Room information is incomplete; system prompts administrator to complete required fields.
5A1	Room number already exists; system rejects request and notifies administrator.
6A1	Invalid room type entered; system requests selection of valid room type (regular/special).
1A1	User is not an administrator; system rejects request and denies access.

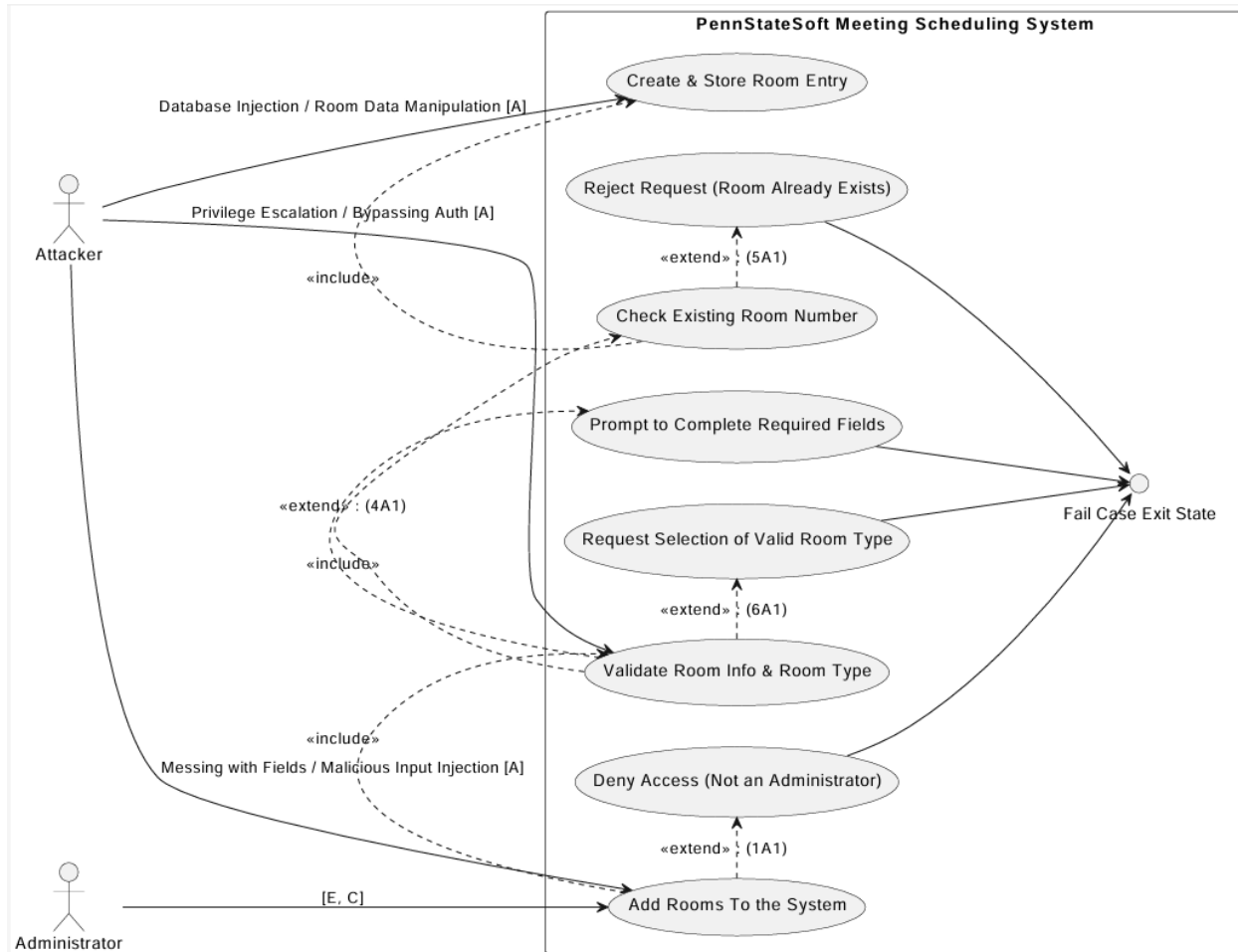


Figure 10.1: Add Meeting Room Use Case Diagram

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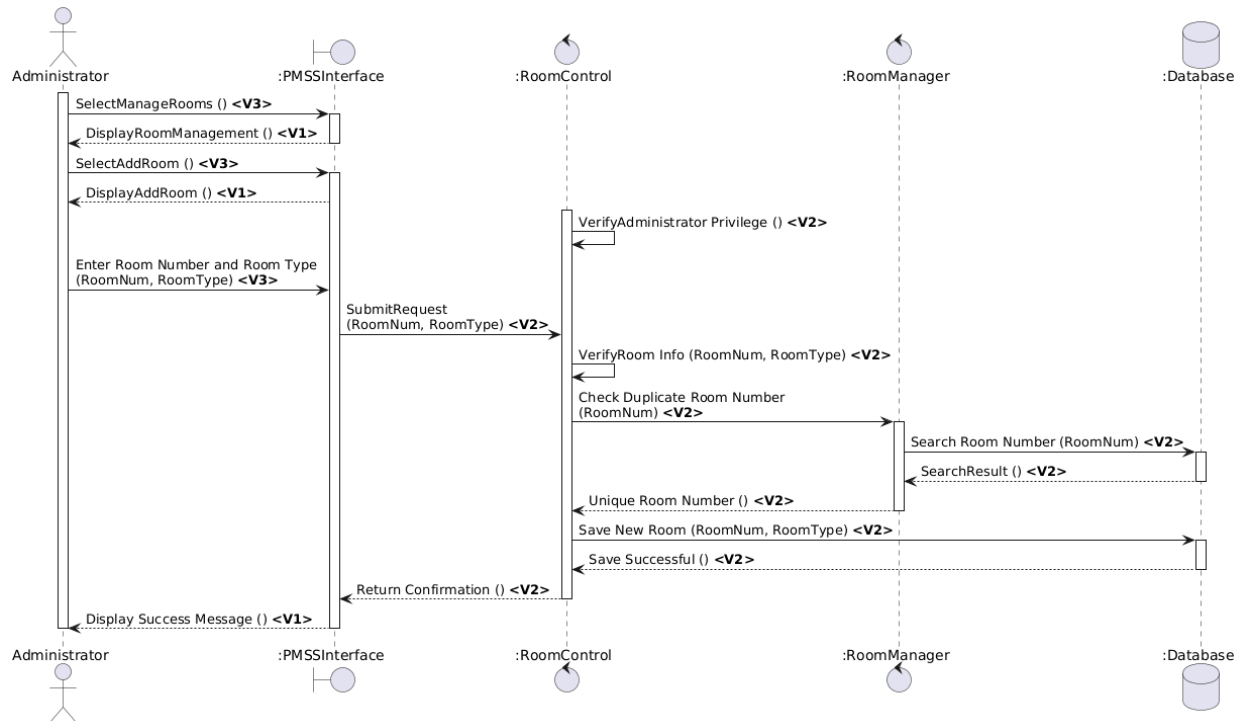


Figure 10.2: Add Meeting Room Sequence Diagram

Use Case ID	Usecase-11
Use Case Name	Delete Meeting Room
User Goal	Allow the administrator to remove a room from the pool of available meeting rooms.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Administrator
Precondition	Administrator is authenticated and logged into the system and the room exists in the pool of available rooms.
Minimal Guarantee	The system does not delete the room if an error occurs or if the room cannot be safely removed. Existing meeting data remains intact.
Success Guarantee	The selected room is removed from the pool of available rooms and is no longer available for future meeting reservations
Fail Case	The administrator requests invalid room (does not exist or schedule conflict); System encounters error.
Consequence of Failure	System rejects modification request and prevents meeting room removal.
Associated Risks	Attackers may manipulate meeting data, alter identifiers and intercept server communication, and gain unauthorized access to database.
Trigger	Administrator selects a room and clicks the “delete room” button.
Steps	
1	System displays the list of available meeting rooms.

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2	Administrator selects the room to delete.
3	Administrator clicks the “Delete Room” button.
4	System checks if the room is currently assigned to a scheduled meeting.
5	System prompts administrator to confirm the deletion
6	Administrator confirms deletion.
7	System removes the room from the pool of available meeting rooms.
8	System displays a confirmation message that the room has been successfully deleted.
Extension Steps	
2A1	Administrator selects a room that does not exist
2A2	System displays an error message and returns administrator to the room management page.
4A1	The selected room is assigned to one or more scheduled meetings.
4A2	System displays an error message indicating that the room cannot be deleted until all associated meetings are canceled or reassigned.
4A3	System cancels room deletion request
6A1	Administrator cancels the deletion confirmation.
6A2	System cancels the deletion request and returns the administrator to the room management page.
7A1	System experiences an error when trying to delete the room.
7A2	System displays an error message requesting the administrator try again.
7A3	System logs error and does not modify room details.

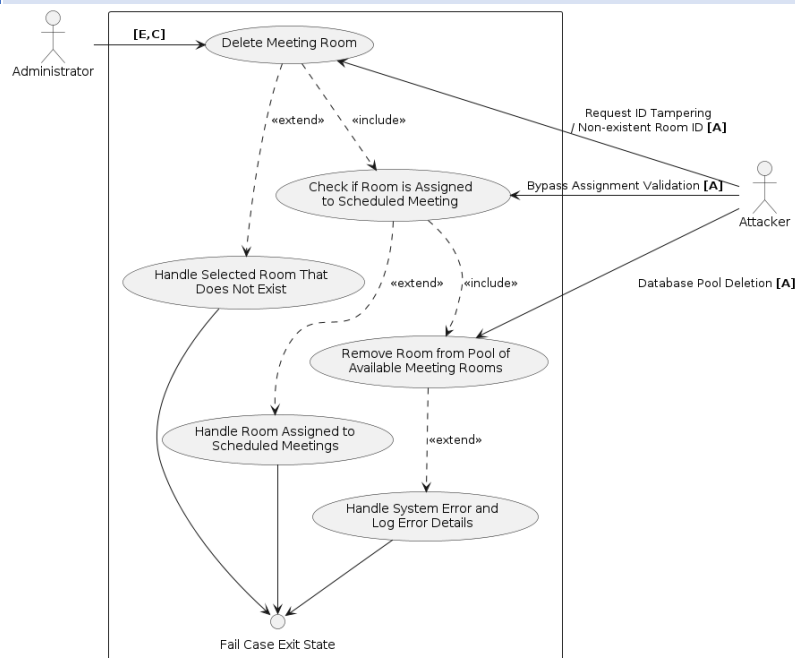


Figure 11.1: Delete Meeting Room Use Case Diagram

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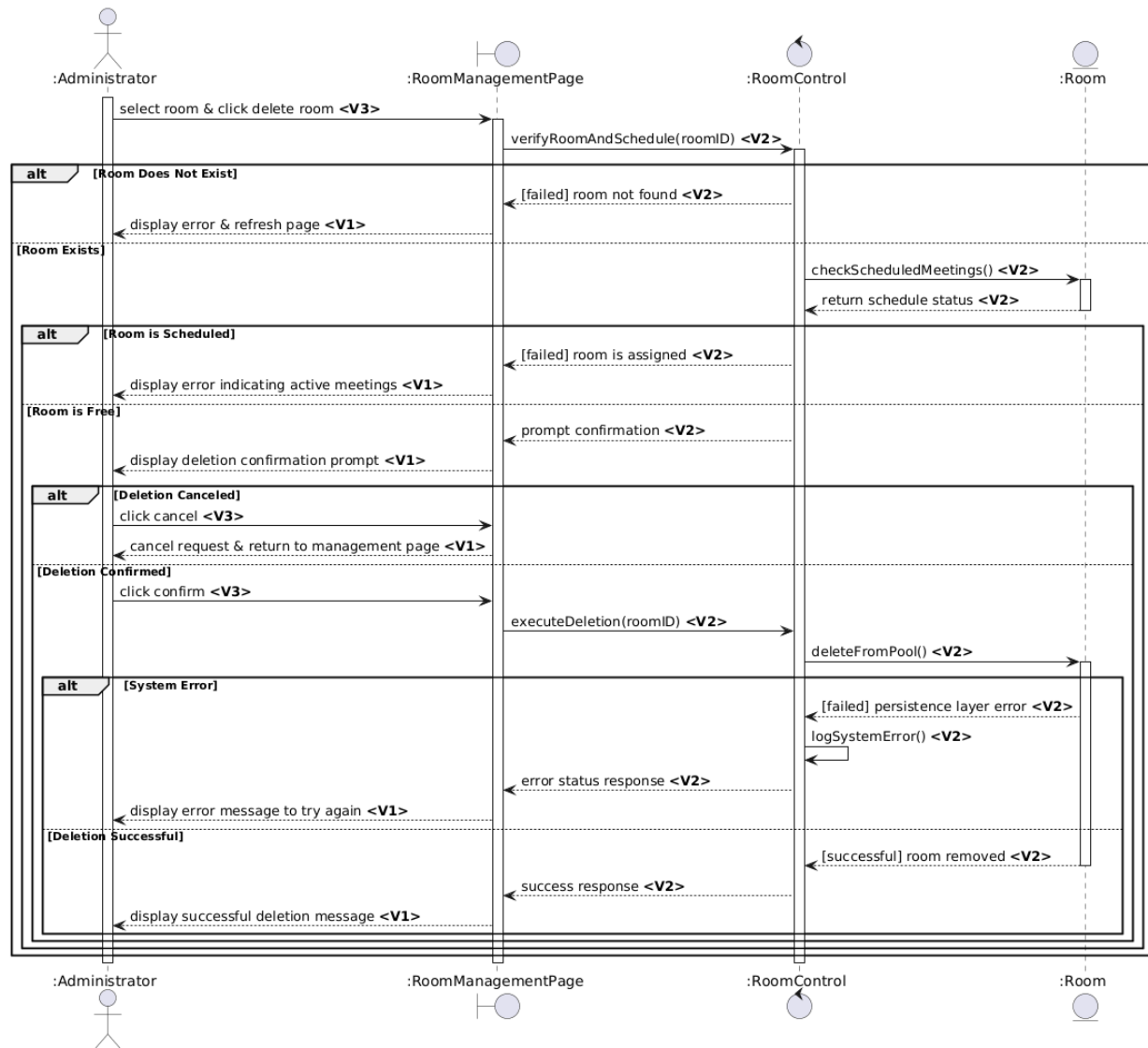


Figure 11.2: Delete Meeting Room Sequence Diagram

Use Case ID	Usecase-12
Use Case Name	View and Edit Client Billing Information
User Goal	Allow the administrator to view and update a client's billing information.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Administrator
Precondition	Administrator is authenticated and logged into the system and client account exists.
Minimal Guarantee	The system does not modify billing information if an error occurs or if the update is cancelled. Existing billing information remains intact.

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Success Guarantee	The administrator can view and update the client's billing information, and the changes are stored securely
Fail Case	The administrator inputs invalid information or fails validation; system encounters error.
Consequence of Failure	The system rejects modification request and preserves original billing information.
Associated Risks	Attacker may gain unauthorized access and extract sensitive data by bypassing restrictions and committing database injection.
Trigger	Administrator selects a client account and clicks the "view/edit billing information" button.
Steps	
1	System displays the list of client accounts.
2	Administrator selects a client account.
3	System displays client current billing information.
4	Administrator edits billing information.
5	Administrator submits the updated billing information.
6	System validates the updated billing information.
7	System securely stores updated billing information.
8	System displays confirmation message indicating billing information has been updated.
Extension Steps	
4A1	Administrator enters invalid/incomplete billing information.
4A2	System displays validation error message and does not submit until all information is corrected.
6A1	The updated billing information fails system validation
6A2	System displays error message and returns to billing information page.
7A1	System experiences an error when trying to save updated billing information
7A2	System displays an error message requesting the administrator try again.
7A3	System logs error and does not modify original billing information.

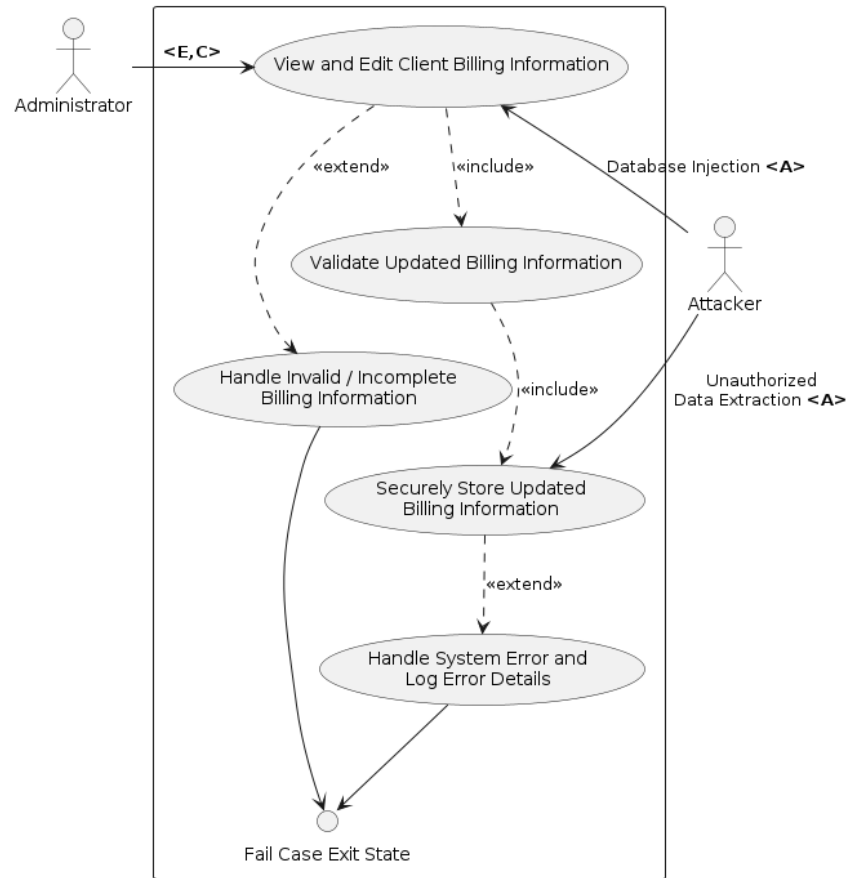


Figure 12.1: Edit Billing Use Case Diagram

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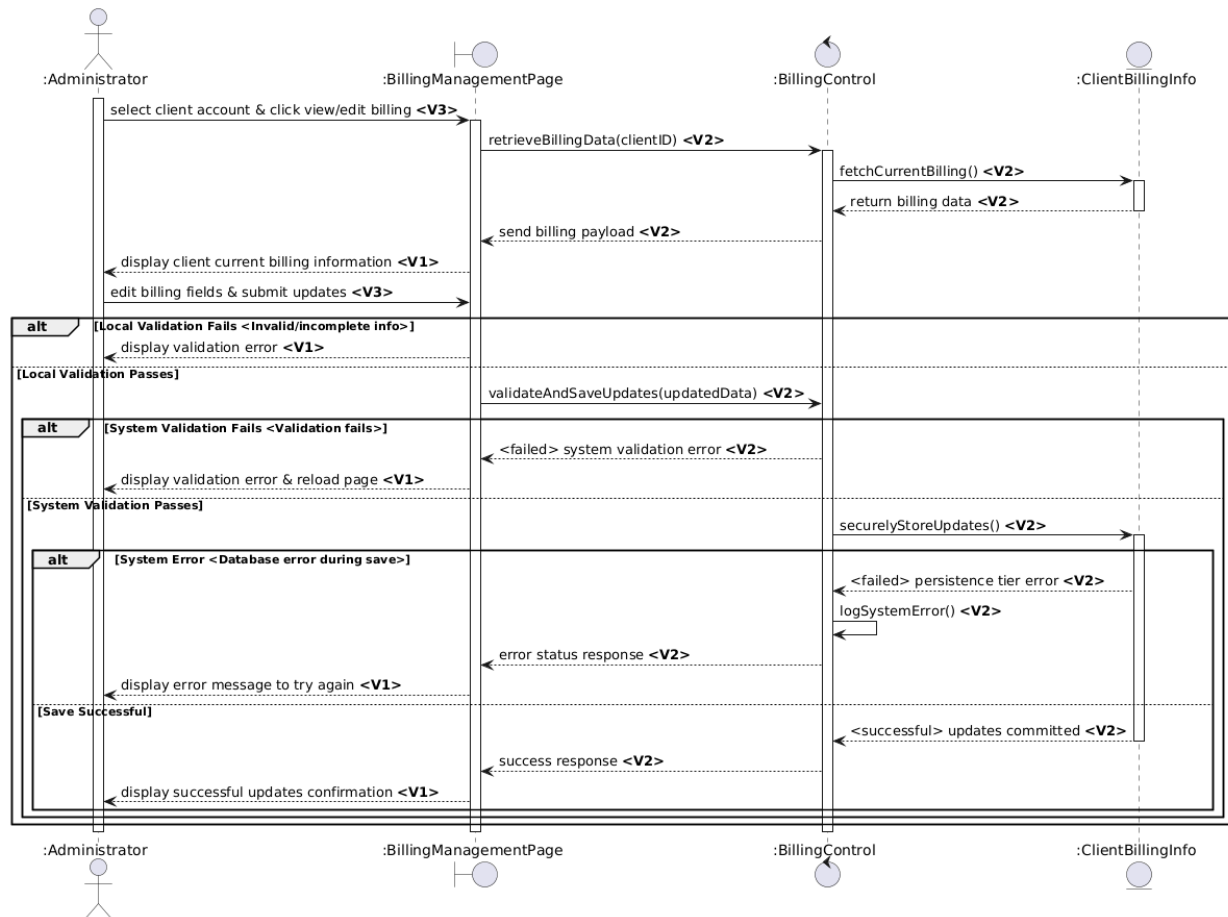


Figure 12.2: Edit Billing Sequence Diagram

Use Case ID	Usecase-13
Use Case Name	View and Respond to Complaints
User Goal	The administrator views client complaints and submits responses to them.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Administrator
Precondition	The administrator is authenticated and has access to the complaint management module.
Minimal Guarantee	No unauthorized access or modification of complaint data occurs; system ensures data integrity even if operation fails.
Success Guarantee	The complaint is successfully reviewed, a response is recorded, and the complaint status is updated in the system.
Fail Case	An unauthorized user accesses complaint records, or an administrator submits invalid or unsafe response content.

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Consequence of Failure	Complaint information may be exposed to unauthorized users, complaint records may be modified incorrectly, or unsafe content may be stored in the system.
Associated Risks	Unauthorized access to complaint information; Improper modification of complaint records; Disclosure of sensitive user information; Invalid or unsafe administrator response content
Trigger	The administrator selects “Complaints” from the system dashboard.
Steps	
1	Administrator selects Complaints.
2	System retrieves and displays the list of submitted complaints.
3	Administrator selects a complaint to review.
4	System displays complaint details.
5	Administrator enters a response.
6	System validates and sanitizes the response content.
7	System stores the administrator response in the database.
8	System updates the complaint status as “Responded”.
9	System displays a confirmation message.
Extension Steps	
2A1	No complaints available; system displays an empty complaint list.
3A1	Selected complaint no longer exists; system shows an error message and refreshes list.
5A1	Administrator submits empty response; system requests valid input.
1A1	User is not an administrator; system rejects access to complaint module.

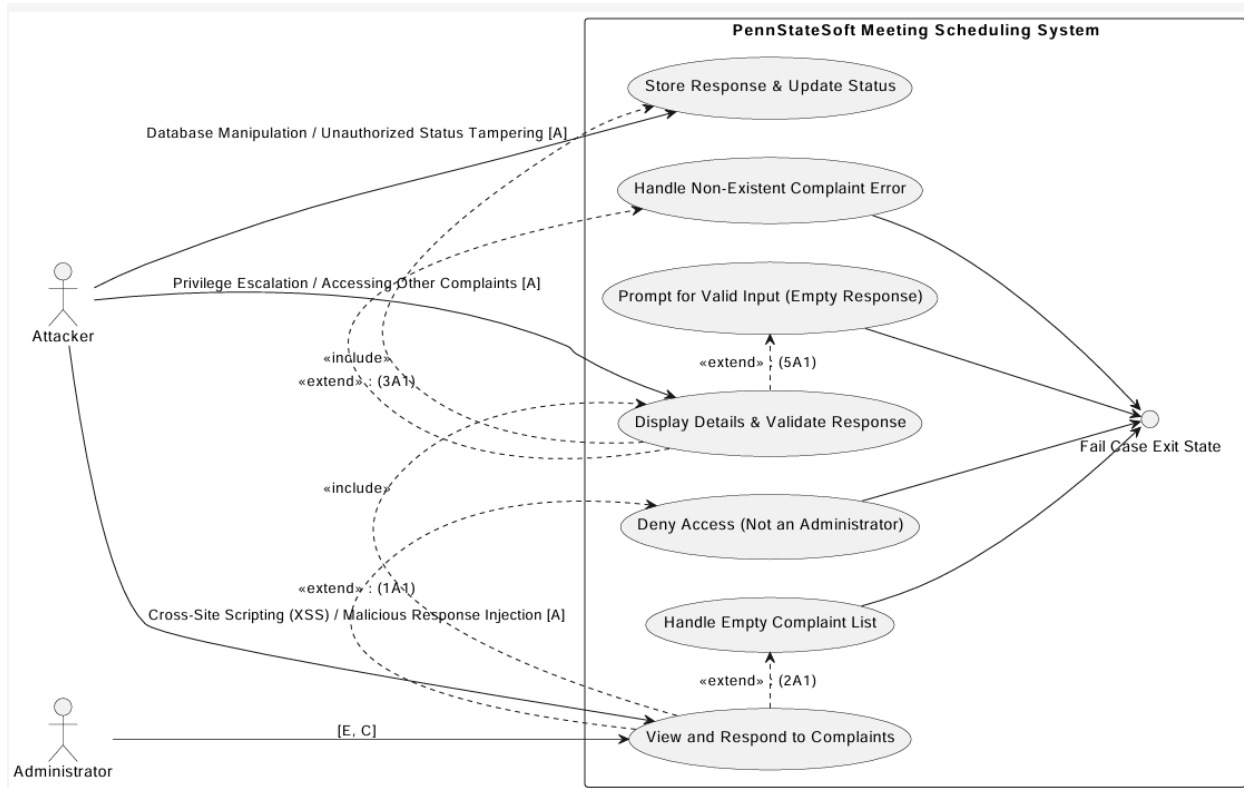


Figure 13.1: Complaint Response Use Case Diagram

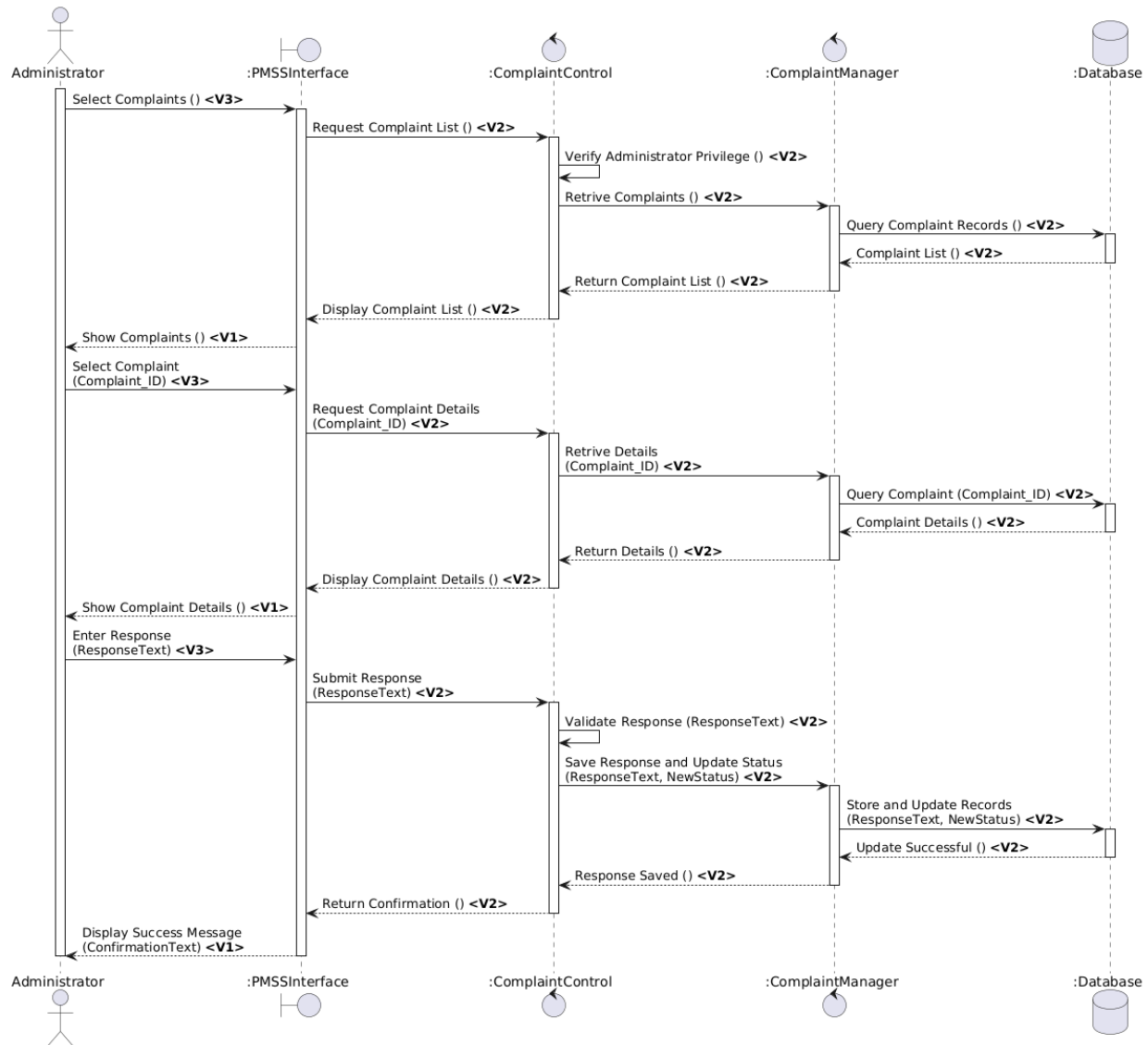


Figure 13.2: Complaint Response Sequence Diagram

Use Case ID	Usecase-14
Use Case Name	Display All Meetings
User Goal	Allow administrator to display all meetings in weekly and daily timeframes.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Administrator
Precondition	The administrator is logged in and has authorization to view all meetings.
Minimal Guarantee	System does not display meetings, returns administrator to homepage.
Success Guarantee	System displays all meetings for administrator in their chosen timeframe.

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Fail Case	Meetings not displayed.
Consequence of Failure	Administrator is unable to view all of the meetings in their desired timeframe.
Associated Risks	Minimal, administrator is just unable to view information of meetings.
Trigger	Administrator clicks “Display All Meetings” button
Steps	
1	The system displays a form asking administrator to choose between viewing all meetings by weekly or daily timeframes.
2	The administrator selects “weekly” and submits the form.
3	The system retrieves all relevant meetings from the week and displays them.
Extension Steps	
2A1	The administrator selects “daily”.
2A2	The system displays a conditional dropdown field prompting the administrator to select their desired day.
2A3	The administrator selects a day from the dropdown and submits the form.
2A4	The system retrieves all meetings and displays them.
2A4A	If no meetings are scheduled for the given day, the system prints “No meetings scheduled.”
3A1	If no meetings are scheduled for the given week, the system prints “No meetings scheduled.”

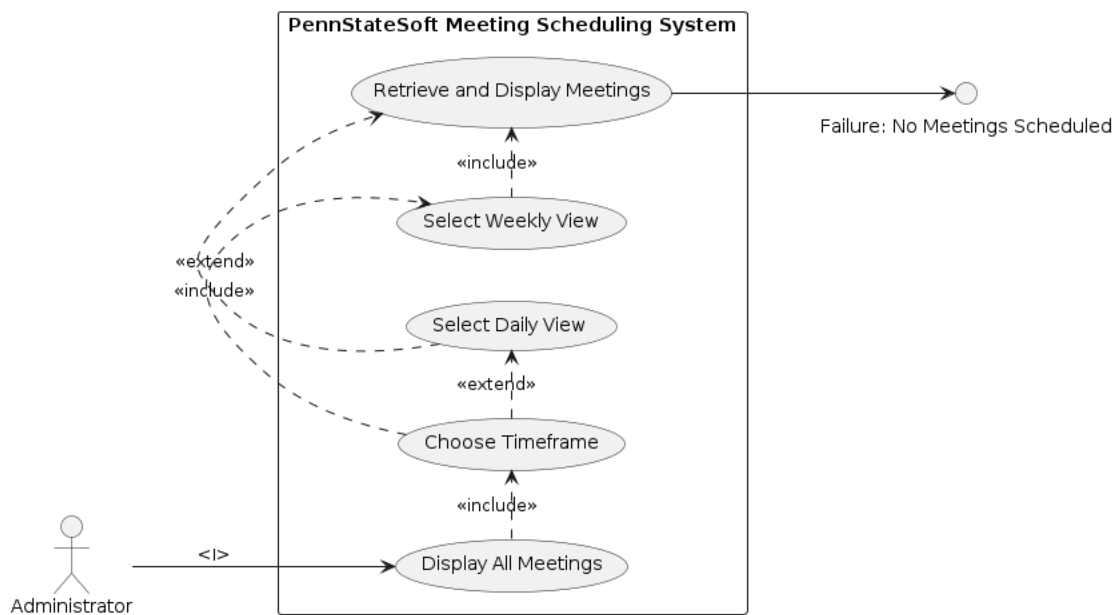


Figure 14.1: Display All Meetings Use Case Diagram

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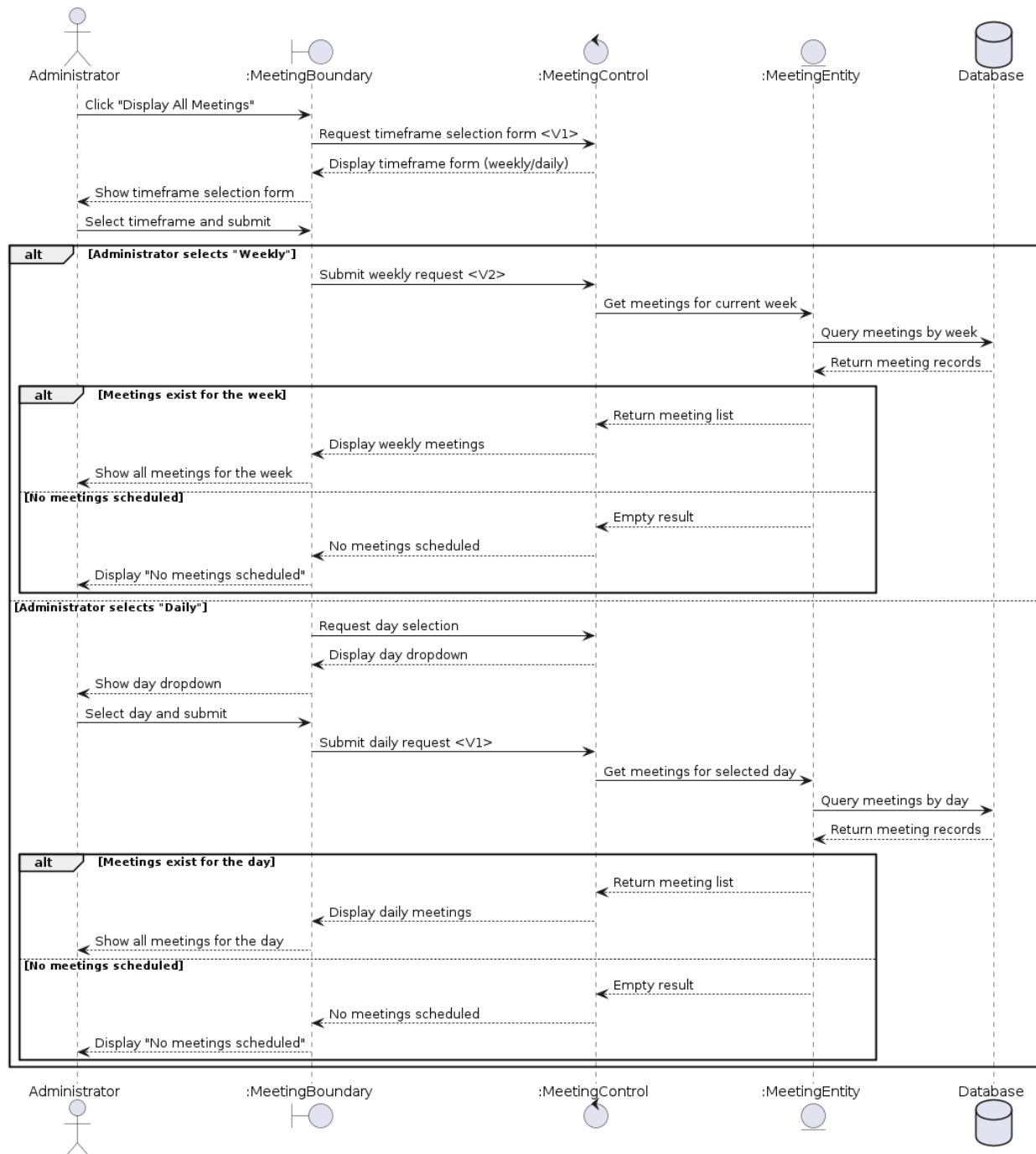


Figure 14.2: Display All Meetings Sequence Diagram

Use Case ID	Usecase-15
Use Case Name	Display All Meetings – Room Specific
User Goal	Allow the administrator to display all meetings for a specific room
Scope	PennStateSoft Meeting Scheduling System

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Level	Primary Task
Primary Actor	Administrator
Precondition	Administrator is logged in and has the authorization to view meetings by room.
Minimal Guarantee	System does not display meetings, returns administrator to the dashboard.
Success Guarantee	System displays all meetings for the administrator for their chosen room.
Fail Case	System does not retrieve the list of meetings for a specific room
Consequence of Failure	Administrator is unable to view the list of meetings for the specified room.
Associated Risks	Unauthorized access to meeting data; Exposure of clients' meeting schedules; Data inconsistency in meeting retrieval by room.
Trigger	Administrator clicks "Display All Meetings" button with room specific search.
Steps	
1	System displays a form asking for the administrator to choose from the available rooms present within the system.
2	Administrator selects a specific room and submits the form.
3	System retrieves all relevant meetings for the specific room.
Extension Steps	
2A1	Administrator selects a specific room.
2A2	System prints "No meetings scheduled." and returns to Administrator dashboard
2B1	Administrator selects a room that is no longer available to be reserved.
2B2	System prints "Room not applicable" and returns to Administrator Dashboard
2C1	Administrator selects a specific room that is a 'special' room.
2C2	System shows all meetings and total amount charged for the week for the specific room.

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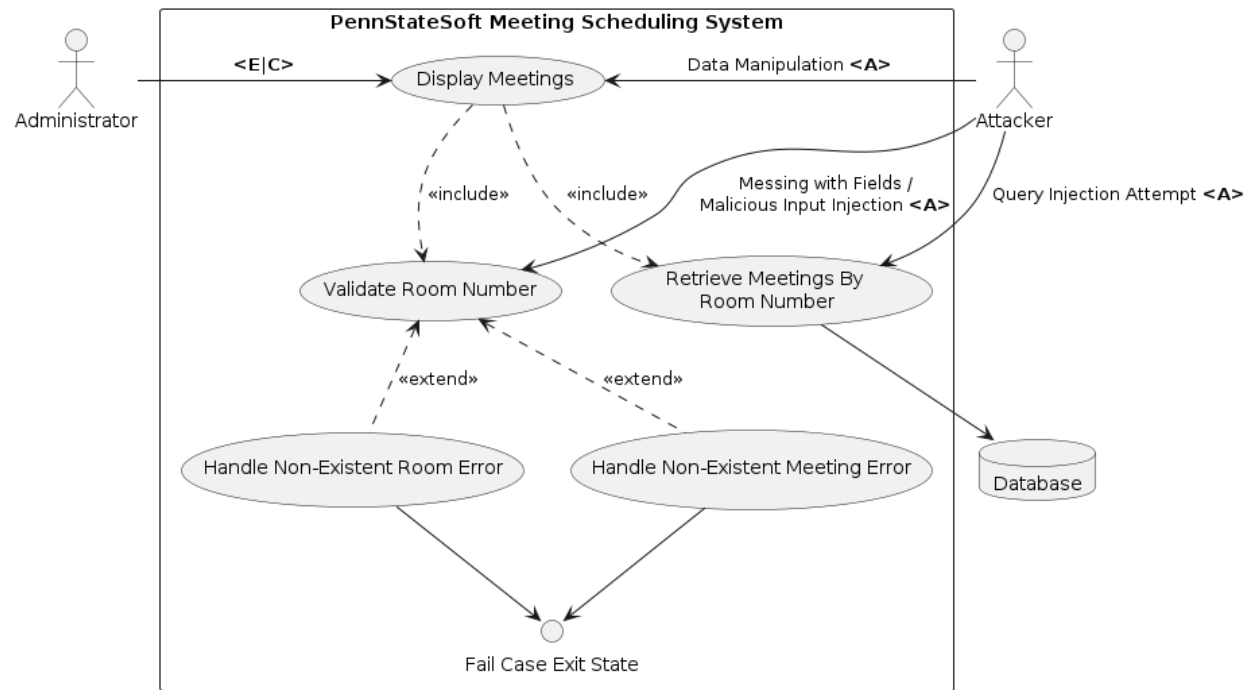


Figure 15.1: Display Room Specific Meetings Use Case Diagram

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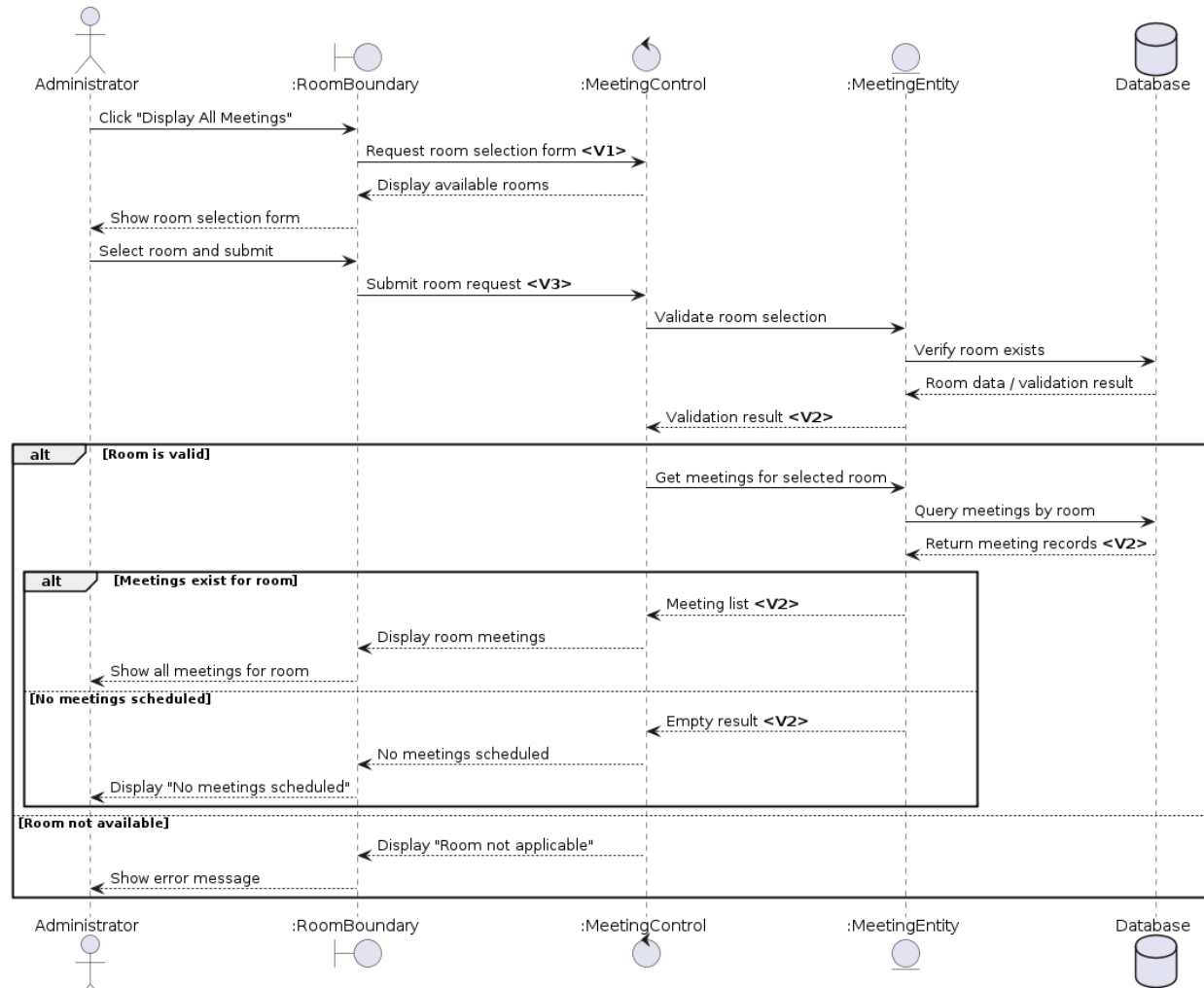


Figure 15.2: Display Room Specific Meetings Sequence Diagram

Use Case ID	Usecase-16
Use Case Name	Display Client's Meetings
User Goal	Allow administrator to view all meetings attended by a specific user.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Administrator
Precondition	Administrator is logged in and has authorization to view meetings of a specific user.
Minimal Guarantee	System does not cause data corruption, returns administrator to dashboard.
Success Guarantee	System displays all meetings of a specific user.
Fail Case	User's meetings not displayed.

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Consequence of Failure	Administrator is unable to view the meetings of a specific user.
Associated Risks	Minimal, administrator is just unable to view the meetings information of a specific user.
Trigger	Administrator clicks “View all of user’s meetings” button on a specific user’s page.
Steps	
1	System displays all meetings attended by user.

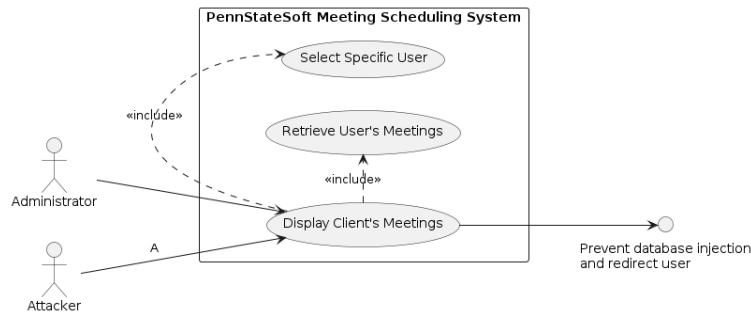


Figure 16.1: Display Client Meetings Use Case Diagram

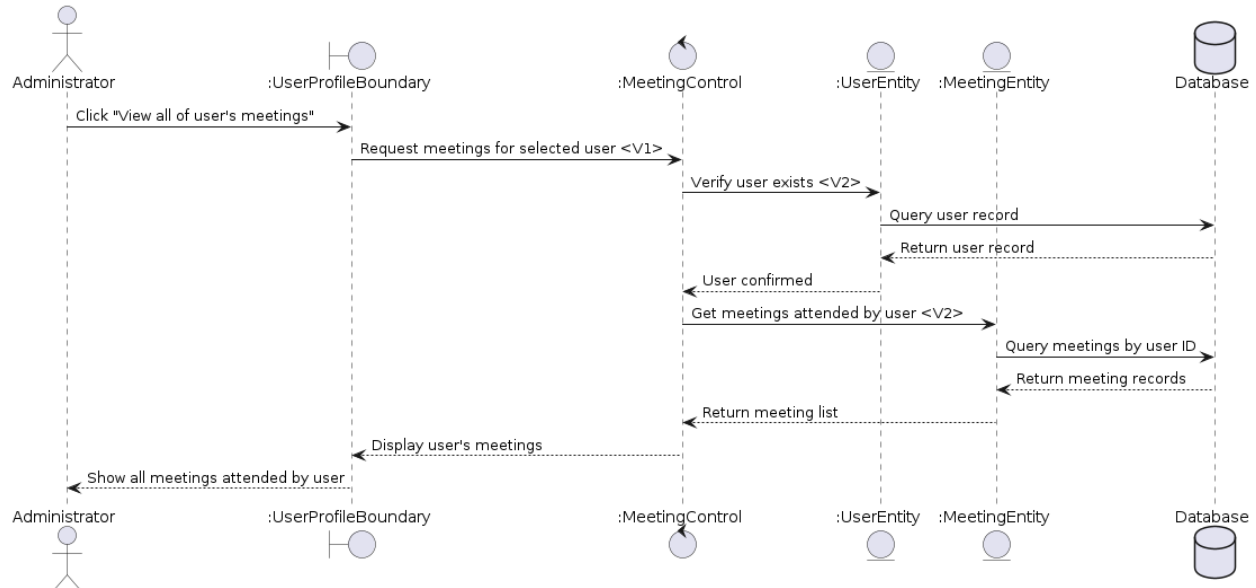


Figure 16.2: Display Client Meetings Sequence Diagram

Use Case ID	Usecase-17
Use Case Name	Display All Meetings – Time Slot Specific
User Goal	Allow the administrator to display all meetings within a specific time slot
Scope	PennStateSoft Meeting Scheduling System

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Level	Primary Task
Primary Actor	Administrator
Precondition	Administrator is logged in and has the authorization to view meetings by room.
Minimal Guarantee	System does not display meetings, returns administrator to the dashboard.
Success Guarantee	System displays all meetings for the administrator for their chosen time slot.
Fail Case	System does not retrieve results for Administrator
Consequence of Failure	Administrator is unable to view meetings for the specific time
Associated Risks	Unauthorized access to meeting data; Exposure of clients' meeting schedules; Data inconsistency in meeting retrieval by time slot
Trigger	Administrator clicks "Display All Meetings" button with time specific search.
Steps	
1	System displays a form asking for the administrator to choose from the available time slot between 9am and 5pm.
2	Administrator selects a specific time period and submits the form.
3	System retrieves all relevant meetings for that specific time period.
Extension Steps	
2A1	Administrator selects a specific time period.
2A2	System prints "No meetings scheduled." and returns to search page
2B1	Administrator selects an invalid timeframe
2B2	System prints "Invalid Time Slot" and returns to search page

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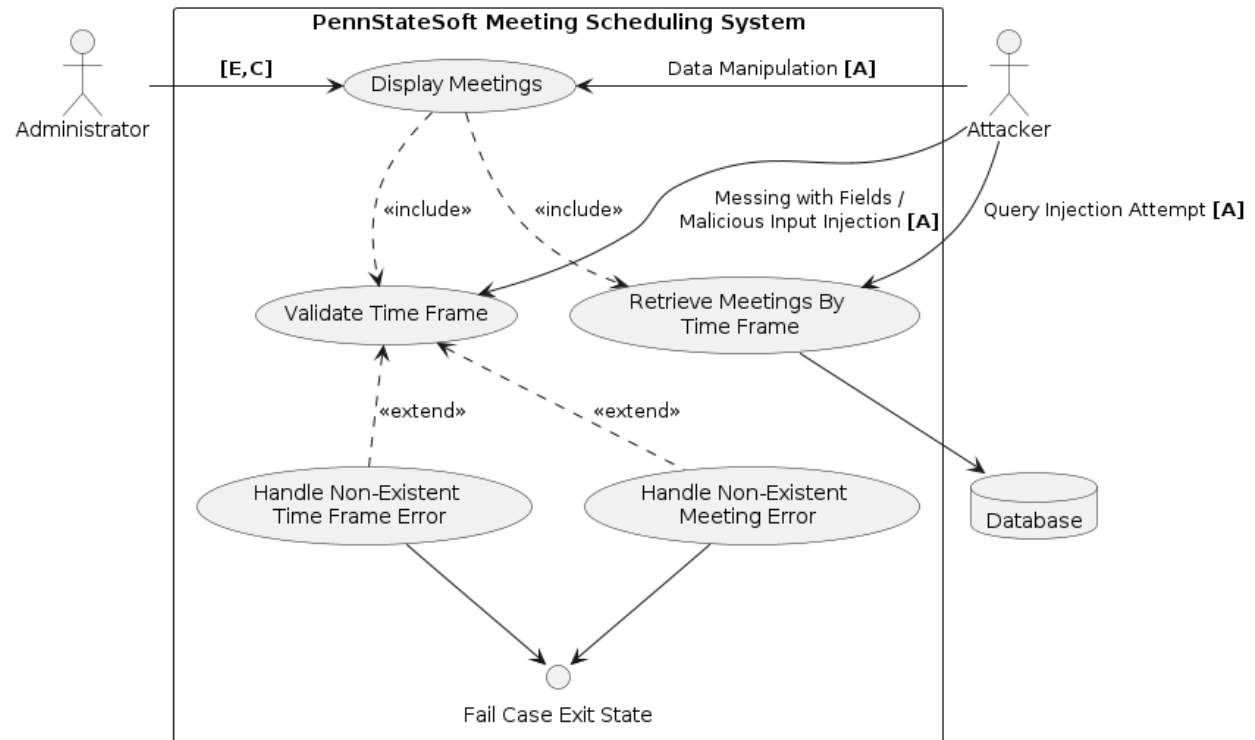


Figure 17.1: Display Time Specific Meetings Use Case Diagram

PennStateSoft	Version: 1.0
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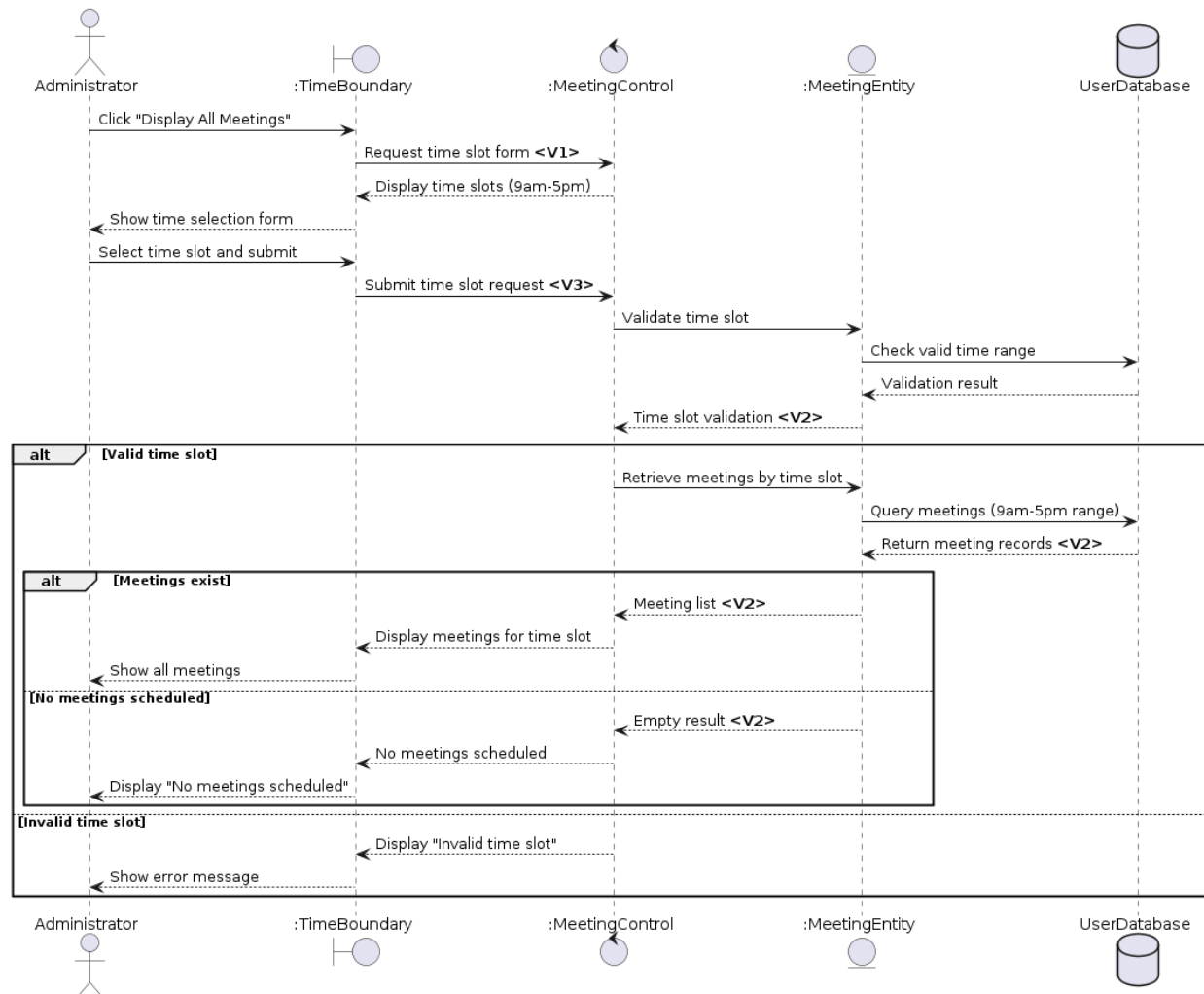


Figure 17.2: Display Time Specific Meetings Sequence Diagram

Use Case ID	Usecase-18
Use Case Name	Create Administrator Account
User Goal	Allow administrator to create a new administrator account.
Scope	PennStateSoft Meeting Scheduling System
Level	Primary Task
Primary Actor	Administrator
Precondition	Administrator is logged in and has authorization to create a new administrator account.
Minimal Guarantee	System does not create new administrator account and redirects user back to their dashboard.
Success Guarantee	Administrator creates a new administrator account.
Fail Case	Administrator account not created.

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Consequence of Failure	Prospective administrator is unable to log in.
Associated Risks	Unauthorized access to system data; Exposure of clients, rooms, and meetings within the system; Data inconsistency in report retrieval
Trigger	Administrator clicks the button to create a new administrator account.
Steps	
1	System displays account registration form.
2	Administrator enters email address into email field.
3	Administrator enters password into password field.
4	Administrator submits account registration form.
5	System verifies no account with given email exists.
6	System creates the new account.
7	System displays success message to administrator and redirects them to the homepage.
Extension Steps	
2A1	Administrator enters invalid email address.
2A2	System displays validation error message and prevents form submission.
3A1	Administrator enters invalid password (e.g. too few characters)
3A2	System displays validation error message and prevents form submission.
5A1	An account with the submitted email already exists.
5A2	System prints an error message informing administrator an account has already registered that email address.
5A3	System cancels execution of account creation.
6A1	System cannot successfully create account.
6A2	System prints the error message informing the administrator of what went wrong.

PennStateSoft	Version: 1.0
	Date: 06/26/26

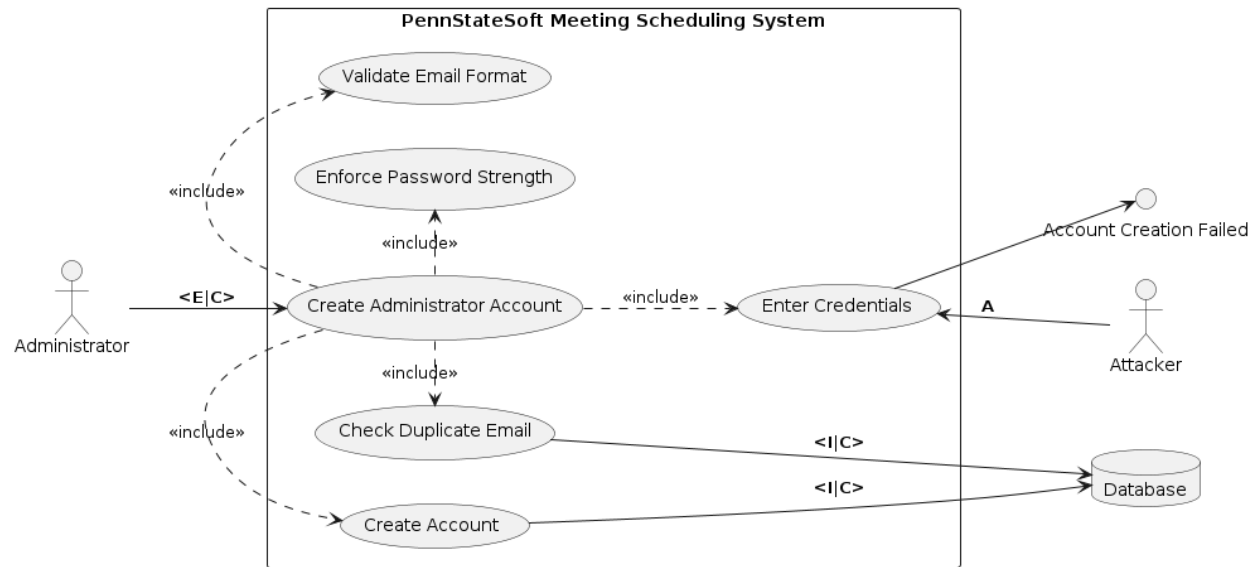


Figure 18.1: Create Administrator Account Use Case Diagram

PennStateSoft	Version: 1.0
	Date: 06/26/26

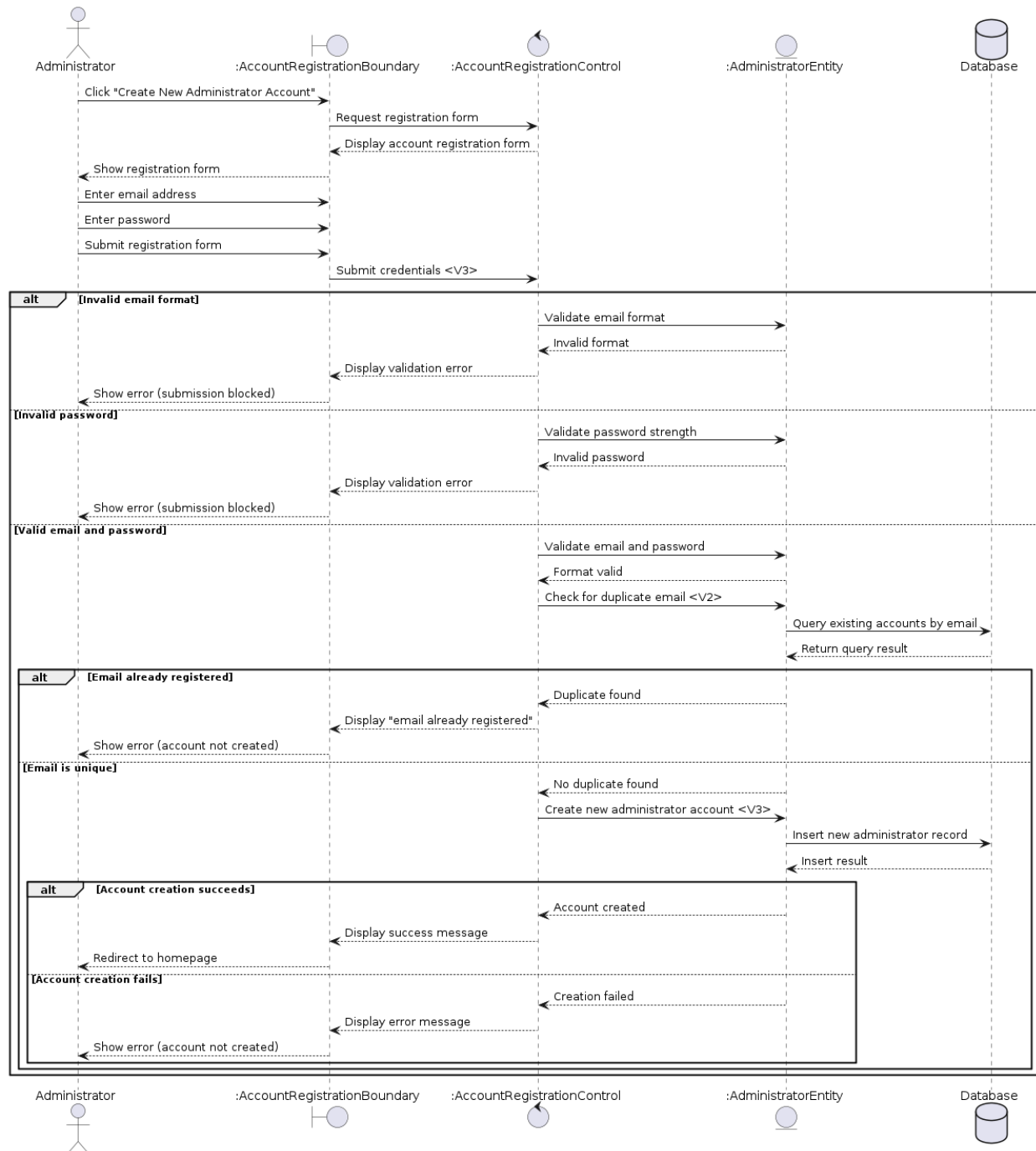


Figure 18.2: Create Administrator Account Sequence Diagram

Use Case ID	Usecase-19
Use Case Name	Generate Reports
User Goal	Allow the administrator to generate reports from the database
Scope	PennStateSoft Meeting Scheduling System

PennStateSoft	Version: 1.0
	Date: 06/26/26

Level	Primary Task
Primary Actor	Administrator
Precondition	Administrator is logged in and has the authorization to generate reports.
Minimal Guarantee	System does not generate reports, returns administrator to the dashboard.
Success Guarantee	System generates reports from the database
Fail Case	Administrator does not have access to reports.
Consequence of Failure	Administrator is unable to generate reports.
Associated Risks	Minimal, administrator may be unable to view reports for the system on rooms, meetings, billing, and clients
Trigger	Administrator clicks “Generate Report” button.
Steps	
1	System displays option for generating report.
2	Administrator selects “Generate Report” button.
3	System pulls reports from the system into a file.
4	System displays access to the file to the Administrator.
Extension Steps	
2A1	Administrator does not have Generate Reports access
2A2	System prints “Access Denied” and returns to Administrator Dashboard
2B1	Administrator selects “Generate Report” twice.
2B2	System prints “Report Already Exists” and returns to Administrator Dashboard

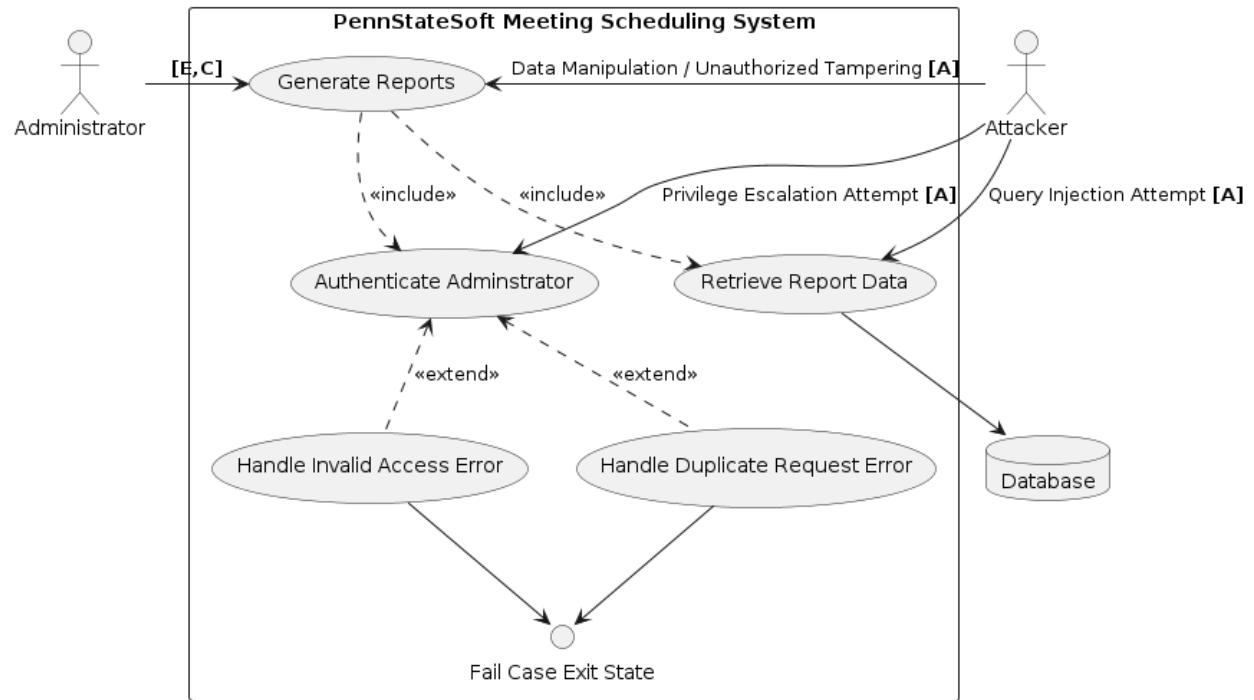


Figure 19.1: Generate Report Use Case Diagram

PennStateSoft	Version: 1.0
	Date: 06/26/26

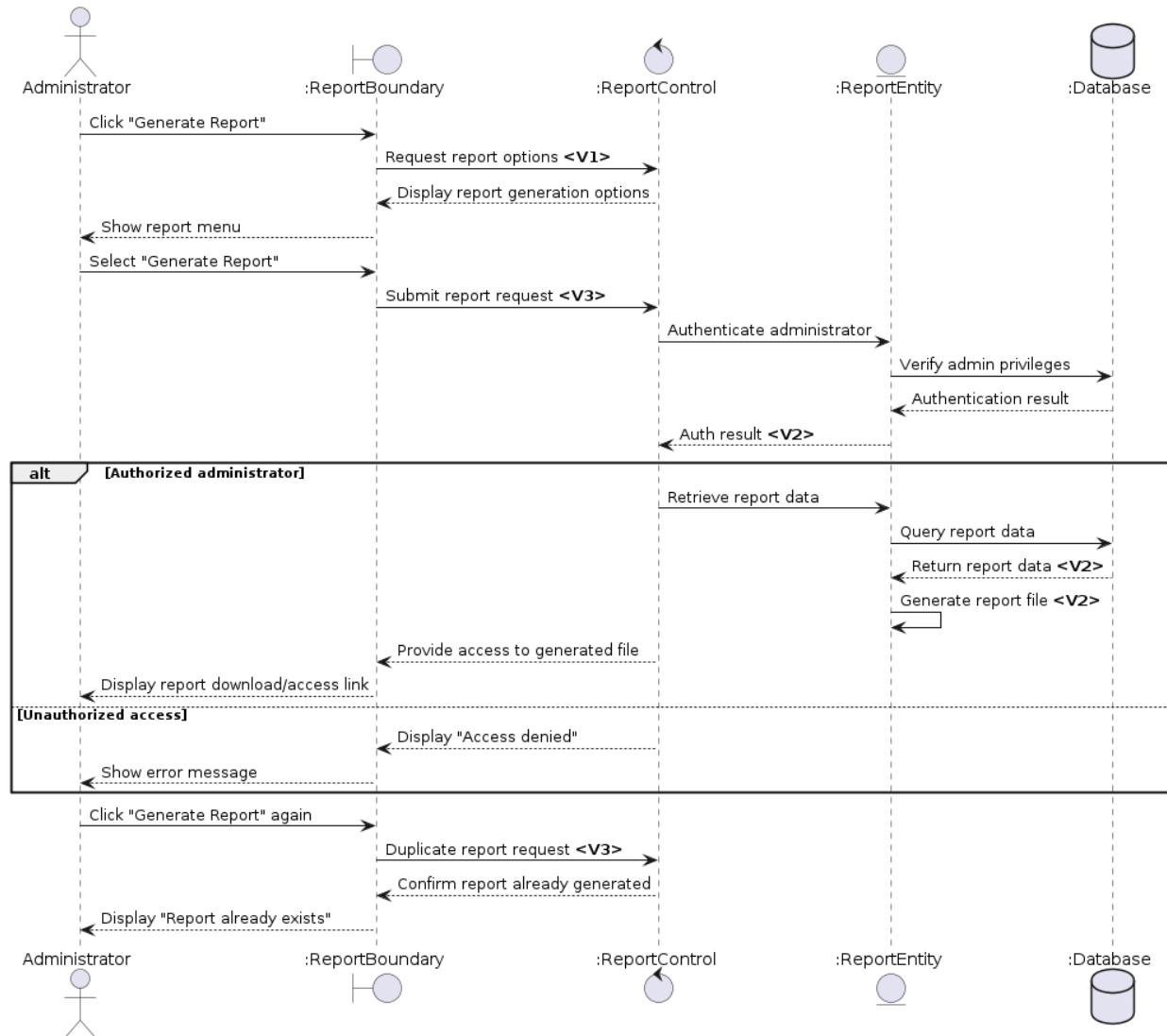


Figure 19.2: Generate Report Sequence Diagram

2.8 Vulnerability Mapping

2.8.1 Use Case - 01: Register Client Account

Step 1: System displays account registration form

Interaction: System to Client UI

Vulnerability Level: V1

Risk: Low risk display only

Mitigation: Safe UI rendering

Step 2: Client enters email address

Interaction: Client to UI (email input)

Vulnerability Level: V3

Risk: Invalid email or fake email input

PennStateSoft	Version: 1.0
	Date: 06/26/26

Mitigation:

Email format validation

Basic input checking

Step 3: Client enters password

Interaction: Client to UI (password input)

Vulnerability Level: V3

Risk: Weak password or invalid password input

Mitigation:

Minimum password length check

Basic password rules

Step 4: Client submits registration form

Interaction: UI to Controller (submit request)

Vulnerability Level: V3

Risk: Malicious request or invalid submission

Mitigation:

HTTPS connection

Form validation before sending

Step 5: System checks if email exists

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Database query manipulation or data leak

Mitigation:

Use safe database query

Check input before query

Step 6: System creates new account

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Duplicate account or incorrect data storage

Mitigation:

Check duplicate email

Ensure safe data storage

Step 7: System shows success message and redirects user

Interaction: Controller to Client UI

Vulnerability Level: V1

Risk: Low risk display only

Mitigation:

PennStateSoft	Version: 1.0
	Date: 06/26/26

Do not show sensitive data

Safe output message

Step 2A1: Invalid email address entered

Vulnerability Level: V3

Risk: Fake or incorrect email input

Mitigation: Email format validation

Step 3A1: Invalid password entered

Vulnerability Level: V3

Risk: Weak password or too short password

Mitigation: Password length check

Step 5A1: Email already exists

Vulnerability Level: V2

Risk: Duplicate account attempt

Mitigation: Check database for existing email before creating account

Step 6A1: System cannot create account

Vulnerability Level: V2

Risk: System failure or database error

Mitigation: Show error message and stop account creation

2.8.2 Use Case - 02: Update Profile

Step 1: System displays home page

Interaction: System to Client UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe UI rendering

Step 2: Client selects view profile

Interaction: Client to UI

Vulnerability Level: V3

Risk: Unauthorized access attempt or invalid request

Mitigation: Check user login session before allowing access

Step 3: System displays profile page

Interaction: System to Client UI

Vulnerability Level: V1

Risk: Low risk display issue

Mitigation: Ensure safe display of user data

PennStateSoft	Version: 1.0
	Date: 06/26/26

Step 4: Client updates profile information

Interaction: Client to UI (profile update input)

Vulnerability Level: V3

Risk: Invalid or inappropriate profile data input

Mitigation:

Input validation

Check allowed values before submission

Step 5: System stores updated profile information

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Incorrect data update or data inconsistency

Mitigation:

Check user authorization

Ensure safe database update

Step 4A1: Client changes display name to nonprofessional name

Vulnerability Level: V3

Risk: Invalid or inappropriate name input

Mitigation: Validate allowed name format

Step 4B1: Client uploads inappropriate profile picture

Vulnerability Level: V3

Risk: Invalid or unsafe image upload

Mitigation: Check file type and image content rules

Step 4C1: Client sets birth date to underage value

Vulnerability Level: V3

Risk: Invalid age information

Mitigation: Validate age requirement before saving profile

2.8.3 Use Case - 03: Create a meeting and reserve a room

Step 1: User selects Create Meeting

Interaction: Client to UI

Vulnerability Level: V3

Risk: Unauthorized or invalid request initiation

Mitigation: Check user login session

Step 2: User enters meeting name, date, and time slot

Interaction: Client to UI (input form)

PennStateSoft	Version: 1.0
	Date: 06/26/26

Vulnerability Level: V3

Risk: Invalid or malicious input data

Mitigation: Input validation for all fields

Step 3: User selects an available room

Interaction: Client to UI

Vulnerability Level: V3

Risk: Selecting invalid or unavailable room

Mitigation: Show only available rooms and validate selection

Step 4: System validates business hours and input completeness

Interaction: Controller processing

Vulnerability Level: V2

Risk: Invalid scheduling rules or incorrect input processing

Mitigation: Enforce business hour rules and validate all input

Step 5: System checks room availability

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Double booking or incorrect room assignment

Mitigation: Use safe database query and locking mechanism

Step 6: System checks client scheduling conflicts

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Client has overlapping meetings

Mitigation: Check schedule conflicts before booking

Step 7: System creates meeting and assigns room

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Incorrect or unauthorized meeting creation

Mitigation: Verify all conditions before database write

Step 8: System stores meeting in database

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data inconsistency or failed storage

Mitigation: Use transaction control for safe saving

Step 9: System displays confirmation message

Interaction: Controller to Client UI

Vulnerability Level: V1

PennStateSoft	Version: 1.0
	Date: 06/26/26

Risk: Low risk display issue only

Mitigation: Safe output display

Step 3A1: No participant is selected

Vulnerability Level: V3

Risk: Missing required input data

Mitigation: Require at least one participant before submission

Step 5A1: Selected room does not exist or is unavailable

Vulnerability Level: V2

Risk: Invalid room assignment

Mitigation: Validate room availability before selection

Step 6A1: Client has scheduling conflicts

Vulnerability Level: V2

Risk: Overlapping meetings for same client

Mitigation: Check schedule before confirming meeting

Step 7A1: Duplicate participants detected

Vulnerability Level: V3

Risk: Duplicate participant entries

Mitigation: Remove duplicates or validate participant list before saving

2.8.4 Use Case - 04: Pay for Special Room Reservation

Step 1: System displays payment page for special room reservations

Interaction: System to Client UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe page rendering

Step 2: Client selects payment method

Interaction: Client to UI

Vulnerability Level: V3

Risk: Invalid or unsafe input selection

Mitigation: Validate allowed payment methods

Step 3: Client enters billing information

Interaction: Client to UI (billing input)

Vulnerability Level: V3

Risk: Invalid or fake billing information

Mitigation: Input validation and format checking

PennStateSoft	Version: 1.0
	Date: 06/26/26

Step 4: Client submits payment request

Interaction: UI to Controller (payment request)

Vulnerability Level: V3

Risk: Unauthorized payment request or request tampering

Mitigation: Use HTTPS and request validation

Step 5: System validates billing information

Interaction: Controller processing

Vulnerability Level: V2

Risk: Incorrect billing data or validation bypass

Mitigation: Validate billing information before processing payment

Step 6: System processes payment through payment service

Interaction: Controller to Payment Service

Vulnerability Level: V2

Risk: Payment service failure or transaction error

Mitigation: Use secure payment API and handle failure safely

Step 7: System confirms successful payment

Interaction: Controller to Client UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe confirmation message display

Step 8: System creates special room reservation

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Room double booking or incorrect reservation

Mitigation: Check room availability before saving and use transaction control

Step 9: System displays reservation confirmation message

Interaction: Controller to Client UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output display

Step 3A1: Client enters invalid or incomplete billing information

Vulnerability Level: V3

Risk: Incorrect billing input

Mitigation: Validate all required billing fields

Step 5A1: Billing information fails validation

Vulnerability Level: V2

PennStateSoft	Version: 1.0
	Date: 06/26/26

Risk: Invalid billing data used in payment

Mitigation: Stop payment process and request correction

Step 6A1: Payment fails due to payment service error

Vulnerability Level: V2

Risk: Payment not completed but process continues

Mitigation: Cancel transaction and ensure rollback

Step 8A1: Special room is no longer available

Vulnerability Level: V2

Risk: Double booking or incorrect reservation

Mitigation: Re-check room availability before final confirmation

2.8.5 Use Case - 05: Add Participants to a Created Meeting

Step 1: User selects one of the meetings they have created

Interaction: Client to UI

Vulnerability Level: V3

Risk: Unauthorized access or invalid meeting selection

Mitigation: Check user login session and meeting ownership

Step 2: User selects Add Participants

Interaction: Client to UI

Vulnerability Level: V3

Risk: Unauthorized action attempt

Mitigation: Verify user permission before allowing access

Step 3: User enters or searches for clients to be added

Interaction: Client to UI (input search)

Vulnerability Level: V3

Risk: Invalid or malicious input or fake client data

Mitigation: Input validation and search filtering

Step 4: System verifies that the user is the creator of the meeting

Interaction: Controller processing

Vulnerability Level: V2

Risk: Unauthorized modification of meeting data

Mitigation: Enforce authorization check before processing

Step 5: System validates that selected clients exist in the system

Interaction: Controller to Database

Vulnerability Level: V2

PennStateSoft	Version: 1.0
	Date: 06/26/26

Risk: Invalid or non existing users added

Mitigation: Validate client records before adding

Step 6: System checks each selected client for scheduling conflicts

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Clients assigned to multiple meetings at same time

Mitigation: Check schedule conflicts before update

Step 7: System verifies that no duplicate participants are added

Interaction: Controller processing

Vulnerability Level: V2

Risk: Duplicate participant records in meeting

Mitigation: Remove duplicates before saving

Step 8: System updates the meeting participant list

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Incorrect or unauthorized data update

Mitigation: Ensure validation is complete before database update

Step 9: System stores the updated meeting information in the database

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data inconsistency or save failure

Mitigation: Use transaction control for safe update

Step 10: System displays a confirmation message to the user

Interaction: Controller to Client UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output display

Step 1A1: User is not the creator of the meeting

Vulnerability Level: V3

Risk: Unauthorized access to modify meeting

Mitigation: Enforce ownership check before allowing operation

Step 3A1: No participant is selected

Vulnerability Level: V3

Risk: Missing required input

Mitigation: Require at least one participant selection

Step 5A1: One or more selected clients do not exist

PennStateSoft	Version: 1.0
	Date: 06/26/26

Vulnerability Level: V2

Risk: Invalid users added to system

Mitigation: Validate user existence before processing

Step 6A1: One or more selected clients have scheduling conflicts

Vulnerability Level: V2

Risk: Schedule overlap for participants

Mitigation: Reject conflicting users and check schedule before adding

Step 7A1: Duplicate participants detected

Vulnerability Level: V2

Risk: Duplicate entries in meeting participant list

Mitigation: Remove duplicates before saving data

2.8.6 Use Case - 06: Remove Participant from Meeting

Step 1: System displays list of all meeting participants

Interaction: System to Client UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe display of participant data

Step 2: Client clicks remove button next to participant

Interaction: Client to UI

Vulnerability Level: V3

Risk: Unauthorized or incorrect removal request

Mitigation: Verify user session and meeting ownership

Step 3: System displays confirmation form

Interaction: System to Client UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe UI rendering

Step 4: Client confirms participant removal

Interaction: Client to UI

Vulnerability Level: V3

Risk: Accidental or unauthorized confirmation action

Mitigation: Require confirmation before processing

Step 5: System removes participant from meeting

Interaction: Controller to Database

PennStateSoft	Version: 1.0
	Date: 06/26/26

Vulnerability Level: V2

Risk: Incorrect data deletion or data inconsistency

Mitigation: Check authorization and ensure valid participant removal

Step 6: System displays success message

Interaction: System to Client UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output display

Step 2A1: Client tries to remove themselves from meeting

Vulnerability Level: V3

Risk: Invalid removal action

Mitigation: Prevent self-removal and show error message

Step 4A1: Client cancels removal

Vulnerability Level: V1

Risk: No real system risk

Mitigation: Simply cancel operation and return to previous page

2.8.7 Use Case - 07: View and edit created meetings

Step 1: System displays list of all meetings created by the client

Interaction: System to Client UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe display of meeting list

Step 2: Client selects a meeting to view

Interaction: Client to UI

Vulnerability Level: V3

Risk: Unauthorized access to meeting details

Mitigation: Check user login session and meeting ownership

Step 3: System displays meeting details

Interaction: System to Client UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe data display

Step 4: Client selects option to edit meeting

Interaction: Client to UI

PennStateSoft	Version: 1.0
	Date: 06/26/26

Vulnerability Level: V3

Risk: Unauthorized edit attempt

Mitigation: Verify user is meeting owner

Step 5: Client updates meeting details

Interaction: Client to UI (input form)

Vulnerability Level: V3

Risk: Invalid or incorrect meeting data input

Mitigation: Input validation for all fields

Step 6: Client submits updated meeting information

Interaction: UI to Controller (submit request)

Vulnerability Level: V3

Risk: Malicious or invalid update request

Mitigation: Validate request before processing

Step 7: System validates change request

Interaction: Controller processing

Vulnerability Level: V2

Risk: Scheduling conflict or invalid update rules

Mitigation: Check room availability and time conflicts

Step 8: System saves meeting changes

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data inconsistency or failed update

Mitigation: Use transaction control and safe database update

Step 9: System displays confirmation message

Interaction: System to Client UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output display

Step 1A1: Client does not have any created meetings

Vulnerability Level: V1

Risk: No system risk

Mitigation: Show empty message safely

Step 5A1: Client provides invalid or incomplete meeting information

Vulnerability Level: V3

Risk: Invalid input data

Mitigation: Validate all input fields before submission

PennStateSoft	Version: 1.0
	Date: 06/26/26

Step 7A1: Meeting time conflicts with another meeting

Vulnerability Level: V2

Risk: Double booking or scheduling conflict

Mitigation: Check time and room availability before saving

Step 7B1: Attendee already scheduled in another meeting

Vulnerability Level: V2

Risk: Participant schedule conflict

Mitigation: Validate participant availability before update

Step 8A1: System error while saving meeting information

Vulnerability Level: V2

Risk: Data not saved or partial update

Mitigation: Use transaction rollback and retry mechanism

2.8.8 Use Case - 08: View planned meetings to attend

Step 1: User selects View My Meetings

Interaction: Client to UI

Vulnerability Level: V3

Risk: Unauthorized or invalid request to access meeting data

Mitigation: Check user login session before allowing access

Step 2: System retrieves the client identity from the active session

Interaction: System processing

Vulnerability Level: V2

Risk: Session hijacking or invalid session usage

Mitigation: Enforce secure session validation and timeout control

Step 3: System queries the database for meetings where the client is a participant

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data leakage or incorrect database query results

Mitigation: Use safe database queries and access control

Step 4: System filters meetings within the current week schedule

Interaction: System processing

Vulnerability Level: V2

Risk: Incorrect filtering or data inconsistency

Mitigation: Apply correct filtering rules and validate results

Step 5: System retrieves meeting details

PennStateSoft	Version: 1.0
	Date: 06/26/26

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Exposure of unauthorized meeting information

Mitigation: Ensure user only accesses own meeting data

Step 6: System displays the list of upcoming meetings

Interaction: System to Client UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output rendering

Step 7: User views meeting details

Interaction: Client to UI

Vulnerability Level: V1

Risk: Low risk viewing action only

Mitigation: No special mitigation required

Step 1A1: Client has no scheduled meetings

Vulnerability Level: V1

Risk: No system risk

Mitigation: Show empty list message safely

Step 2A1: Session has expired

Vulnerability Level: V3

Risk: Unauthorized access attempt

Mitigation: Require re-login and session validation

Step 3A1: Database query fails

Vulnerability Level: V2

Risk: System error or incomplete data retrieval

Mitigation: Handle error safely and allow retry mechanism

2.8.9 Use Case - 09: File a complaint

Step 1: Client clicks the file complaint button

Interaction: Client to UI

Vulnerability Level: V3

Risk: Unauthorized or invalid request to access complaint system

Mitigation: Check user login session before allowing access

Step 2: Client chooses the meeting the complaint is regarding

Interaction: Client to UI

PennStateSoft	Version: 1.0
	Date: 06/26/26

Vulnerability Level: V3

Risk: Selecting invalid meeting or unauthorized meeting access

Mitigation: Validate that user attended the meeting before allowing selection

Step 3: Client chooses from predefined complaint options

Interaction: Client to UI

Vulnerability Level: V3

Risk: Invalid or missing complaint category selection

Mitigation: Enforce selection from predefined valid options

Step 4: Client enters complaint summary

Interaction: Client to UI (text input)

Vulnerability Level: V3

Risk: Malicious or inappropriate text input

Mitigation: Input validation and basic text sanitization

Step 5: Client submits complaint

Interaction: UI to Controller (submit request)

Vulnerability Level: V3

Risk: Invalid or malicious submission request

Mitigation: Validate full form before submission and use HTTPS

Step 6: System saves complaint to database

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data corruption or incorrect storage of complaint

Mitigation: Use safe database insertion and check data integrity

Step 2A1: Client has not attended any meetings

Vulnerability Level: V1

Risk: No system security risk

Mitigation: Show message and disable complaint option

Step 2A2: Complaint submitted without selecting meeting

Vulnerability Level: V3

Risk: Invalid complaint submission

Mitigation: Require meeting selection before enabling submit

Step 3A1: Client does not choose complaint option

Vulnerability Level: V3

Risk: Incomplete complaint data

Mitigation: Require selection of complaint category before submission

PennStateSoft	Version: 1.0
	Date: 06/26/26

2.8.10 Use Case - 10: Add rooms to the system

Step 1: Administrator selects Manage Rooms

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Unauthorized access attempt to room management

Mitigation: Verify administrator login session and role

Step 2: Administrator selects Add Room

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Unauthorized action to modify system resources

Mitigation: Check admin privileges before allowing access

Step 3: Administrator enters room number and room type

Interaction: Admin to UI (input form)

Vulnerability Level: V3

Risk: Invalid or malicious room data input

Mitigation: Input validation for room number and room type

Step 4: System validates room information

Interaction: Controller processing

Vulnerability Level: V2

Risk: Incorrect or incomplete data processing

Mitigation: Validate all required fields before processing

Step 5: System checks whether room number already exists

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Duplicate room records or data inconsistency

Mitigation: Use database uniqueness check for room number

Step 6: System verifies room type

Interaction: Controller processing

Vulnerability Level: V2

Risk: Invalid room category stored in system

Mitigation: Restrict room type to predefined values only

Step 7: System creates new room entry

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Unauthorized or incorrect data insertion

Mitigation: Ensure validation is completed before database write

PennStateSoft	Version: 1.0
	Date: 06/26/26

Step 8: System stores room information in database

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data corruption or failed storage

Mitigation: Use transaction control for safe storage

Step 9: System displays confirmation message

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output display

Step 4A1: Room information is incomplete

Vulnerability Level: V3

Risk: Missing required input data

Mitigation: Require all fields before submission

Step 5A1: Room number already exists

Vulnerability Level: V2

Risk: Duplicate room entry

Mitigation: Check uniqueness before saving

Step 6A1: Invalid room type entered

Vulnerability Level: V3

Risk: Incorrect room category input

Mitigation: Restrict selection to valid room types (regular/special)

Step 1A1: User is not an administrator

Vulnerability Level: V3

Risk: Unauthorized access to system functions

Mitigation: Enforce role-based access control (admin only)

2.8.11 Use Case - 11: Delete meeting rooms

Step 1: System displays list of available meeting rooms

Interaction: System to Administrator UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe UI rendering of room list

Step 2: Administrator selects a room to delete

Interaction: Admin to UI

PennStateSoft	Version: 1.0
	Date: 06/26/26

Vulnerability Level: V3

Risk: Unauthorized or incorrect room selection

Mitigation: Check administrator login session and role

Step 3: Administrator clicks Delete Room button

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Unauthorized delete request

Mitigation: Verify admin privileges before allowing action

Step 4: System checks if room is assigned to scheduled meetings

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Deleting room that is in use causing data inconsistency

Mitigation: Check room booking before allowing deletion

Step 5: System prompts administrator to confirm deletion

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe confirmation prompt

Step 6: Administrator confirms deletion

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Accidental or unauthorized confirmation

Mitigation: Require confirmation step before deletion

Step 7: System removes room from database

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data inconsistency or failed deletion

Mitigation: Use transaction control and safe delete operation

Step 8: System displays confirmation message

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output display

Step 2A1: Administrator selects a room that does not exist

Vulnerability Level: V3

Risk: Invalid deletion request

PennStateSoft	Version: 1.0
	Date: 06/26/26

Mitigation: Validate room existence before processing

Step 4A1: Room is assigned to scheduled meetings

Vulnerability Level: V2

Risk: Breaking existing meeting data

Mitigation: Prevent deletion until meetings are reassigned or canceled

Step 6A1: Administrator cancels deletion

Vulnerability Level: V1

Risk: No system risk

Mitigation: Simply stop process and return to room management page

Step 7A1: System error during deletion

Vulnerability Level: V2

Risk: Partial deletion or data inconsistency

Mitigation: Use rollback and error handling, log the issue

2.8.12 Use Case - 12: View and edit client billing information

Step 1: System displays list of client accounts

Interaction: System to Administrator UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe UI rendering of account list

Step 2: Administrator selects a client account

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Unauthorized access or incorrect account selection

Mitigation: Check administrator authentication and authorization

Step 3: System displays client billing information

Interaction: System to Admin UI

Vulnerability Level: V2

Risk: Exposure of sensitive billing data

Mitigation: Restrict billing data access to administrators only

Step 4: Administrator edits billing information

Interaction: Admin to UI (billing input)

Vulnerability Level: V3

Risk: Invalid or malicious billing data input

Mitigation: Input validation for billing fields

PennStateSoft	Version: 1.0
	Date: 06/26/26

Step 5: Administrator submits updated billing information

Interaction: UI to Controller (submit request)

Vulnerability Level: V3

Risk: Unauthorized or incorrect update request

Mitigation: Validate request and admin session before processing

Step 6: System validates updated billing information

Interaction: Controller processing

Vulnerability Level: V2

Risk: Incorrect or incomplete billing data

Mitigation: Validate all billing fields before saving

Step 7: System stores updated billing information

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data inconsistency or failed update

Mitigation: Use secure database update and transaction control

Step 8: System displays confirmation message

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output display

Step 4A1: Administrator enters invalid or incomplete billing information

Vulnerability Level: V3

Risk: Incorrect billing data input

Mitigation: Require complete and valid billing information before submission

Step 6A1: Billing information fails validation

Vulnerability Level: V2

Risk: Invalid data stored or rejected update

Mitigation: Stop update and request corrected data

Step 7A1: System error while saving billing information

Vulnerability Level: V2

Risk: Data not saved or inconsistent update

Mitigation: Use rollback mechanism and log error, keep original data unchanged

2.8.13 Use Case - 13: View and respond to complaints

Step 1: Administrator selects Complaints

PennStateSoft	Version: 1.0
	Date: 06/26/26

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Unauthorized access to complaint system

Mitigation: Check administrator login session and role

Step 2: System retrieves and displays list of complaints

Interaction: System to Admin UI

Vulnerability Level: V2

Risk: Exposure of sensitive complaint data

Mitigation: Restrict complaint data access to administrators only

Step 3: Administrator selects a complaint

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Invalid or unauthorized complaint access

Mitigation: Validate complaint existence and admin access rights

Step 4: System displays complaint details

Interaction: System to Admin UI

Vulnerability Level: V2

Risk: Leakage of sensitive user information

Mitigation: Ensure only authorized admin can view complaint details

Step 5: Administrator enters response

Interaction: Admin to UI (text input)

Vulnerability Level: V3

Risk: Malicious or invalid text input

Mitigation: Input validation and basic sanitization

Step 6: System validates response content

Interaction: Controller processing

Vulnerability Level: V2

Risk: Inappropriate or unsafe response content

Mitigation: Validate and sanitize administrator input

Step 7: System stores administrator response

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data corruption or failed storage

Mitigation: Use safe database write and validation checks

Step 8: System updates complaint status

Interaction: Controller to Database

PennStateSoft	Version: 1.0
	Date: 06/26/26

Vulnerability Level: V2

Risk: Incorrect status update or data inconsistency

Mitigation: Ensure transaction consistency before update

Step 9: System displays confirmation message

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output display

Step 2A1: No complaints available

Vulnerability Level: V1

Risk: No security risk

Mitigation: Show empty complaint list safely

Step 3A1: Selected complaint no longer exists

Vulnerability Level: V2

Risk: Data inconsistency or invalid access

Mitigation: Refresh complaint list and validate selection

Step 5A1: Administrator submits empty response

Vulnerability Level: V3

Risk: Invalid response input

Mitigation: Require valid response before submission

Step 1A1: User is not an administrator

Vulnerability Level: V3

Risk: Unauthorized access to complaint system

Mitigation: Enforce role-based access control (admin only)

2.8.14 Use Case - 14: Display all meetings

Step 1: System displays form for selecting weekly or daily timeframe

Interaction: System to Administrator UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe UI rendering

Step 2: Administrator selects weekly timeframe and submits form

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Unauthorized request or invalid selection input

PennStateSoft	Version: 1.0
	Date: 06/26/26

Mitigation: Check administrator authentication and validate input selection

Step 3: System retrieves all meetings for the week

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data exposure or incorrect data retrieval

Mitigation: Restrict access to authorized administrators only and use safe queries

Step 4: System displays all weekly meetings

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output display

Step 2A1: Administrator selects daily option

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Invalid or manipulated input selection

Mitigation: Validate allowed options (weekly or daily only)

Step 2A2: System displays dropdown for selecting day

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: Low risk UI issue only

Mitigation: Safe UI rendering

Step 2A3: Administrator selects a day and submits

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Invalid day selection or unauthorized request

Mitigation: Validate selected day input

Step 2A4: System retrieves meetings for selected day

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data leakage or incorrect query results

Mitigation: Use safe database query and access control

Step 2A4A: No meetings scheduled for selected day

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: No security risk

Mitigation: Display safe message only

PennStateSoft	Version: 1.0
	Date: 06/26/26

Step 3A1: No meetings scheduled for selected week

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: No security risk

Mitigation: Display safe message only

2.8.15 Use Case - 15: Display All Meetings – Room Specific

Step 1: System displays form for selecting available rooms

Interaction: System to Administrator UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe UI rendering of room list

Step 2: Administrator selects a specific room and submits form

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Unauthorized or invalid room selection

Mitigation: Check administrator authentication and validate room selection

Step 3: System retrieves meetings for selected room

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data exposure or incorrect query results

Mitigation: Use safe database query and restrict access to admin only

Step 4: System displays meetings for selected room

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output rendering

Step 2A1: Administrator selects a valid room

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Invalid or unauthorized room access attempt

Mitigation: Validate room existence and admin access rights

Step 2B1: Administrator selects a room that is no longer available

Interaction: System to Admin UI

Vulnerability Level: V2

PennStateSoft	Version: 1.0
	Date: 06/26/26

Risk: Invalid data access or inconsistent system state

Mitigation: Check room status before processing request

Step 2C1: Administrator selects a special room

Interaction: System processing

Vulnerability Level: V2

Risk: Exposure of financial or sensitive usage data

Mitigation: Restrict access and ensure safe calculation of room usage data

2.8.16 Use Case - 16: Display Client's Meetings

Step 1: System displays all meetings attended by user

Interaction: System to Admin UI

Vulnerability Level: V2

Risk: Sensitive user meeting data may be exposed without proper control

Mitigation: Restrict access to authorized administrators only and validate user permission before query

Step 2: Administrator requests to view user meetings

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Unauthorized access to another user's meeting data

Mitigation: Check administrator authentication and authorization before processing request

Step 3: System retrieves user meeting data from database

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data leakage or incorrect query results

Mitigation: Use safe database query and ensure access control rules are applied

Step 4: System displays meeting information

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output rendering and proper data formatting

Step 1A1: User has no meetings

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: No security risk

Mitigation: Display safe empty message only

PennStateSoft	Version: 1.0
	Date: 06/26/26

2.8.17 Use Case - 17: Display All Meetings – Time Slot Specific

Step 1: System displays form for selecting time slot between 9am and 5pm

Interaction: System to Administrator UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe UI rendering of time slot selection

Step 2: Administrator selects a specific time period and submits form

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Unauthorized or invalid time slot selection

Mitigation: Check administrator authentication and validate selected time range

Step 3: System retrieves meetings for selected time slot

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data exposure or incorrect query results

Mitigation: Use safe database queries and enforce access control

Step 4: System displays meetings for selected time slot

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output rendering

Step 2A1: Administrator selects a specific time period

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Invalid or manipulated time selection

Mitigation: Validate allowed time slots (9am–5pm only)

Step 2A2: No meetings scheduled for selected time period

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: No security risk

Mitigation: Display safe empty message only

2.8.18 Use Case - 18: Create Administrator Account

Step 1: System displays account registration form

Interaction: System to Administrator UI

Vulnerability Level: V1

PennStateSoft	Version: 1.0
	Date: 06/26/26

Risk: Low risk display issue only

Mitigation: Safe UI rendering

Step 2: Administrator enters email address

Interaction: Admin to UI (email input)

Vulnerability Level: V3

Risk: Invalid or fake email input

Mitigation: Email format validation and basic input checking

Step 3: Administrator enters password

Interaction: Admin to UI (password input)

Vulnerability Level: V3

Risk: Weak password or invalid password input

Mitigation: Minimum password length check and password rules enforcement

Step 4: Administrator submits registration form

Interaction: UI to Controller (submit request)

Vulnerability Level: V3

Risk: Malicious or invalid account creation request

Mitigation: Use HTTPS and validate request before processing

Step 5: System verifies email does not already exist

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Duplicate administrator account or data conflict

Mitigation: Use database uniqueness check and safe query

Step 6: System creates new administrator account

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Unauthorized or incorrect account creation

Mitigation: Ensure proper authorization and validate data before saving

Step 7: System displays success message

Interaction: System to Administrator UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe output display

Step 2A1: Administrator enters invalid email address

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Invalid email input

PennStateSoft	Version: 1.0
	Date: 06/26/26

Mitigation: Validate email format before submission

Step 3A1: Administrator enters invalid password

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Weak or insecure password

Mitigation: Enforce password strength rules

Step 5A1: Account already exists with same email

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Duplicate administrator account creation attempt

Mitigation: Check existing email before creating account

Step 6A1: System cannot create account

Interaction: Controller to Database

Vulnerability Level: V2

Risk: System error or failure during account creation

Mitigation: Show error message and prevent partial data creation

2.8.19 Use Case - 19: Generate reports

Step 1: System displays option for generating report

Interaction: System to Administrator UI

Vulnerability Level: V1

Risk: Low risk display issue only

Mitigation: Safe UI rendering

Step 2: Administrator clicks Generate Report button

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Unauthorized or repeated report request

Mitigation: Check administrator authentication and prevent duplicate submission

Step 3: System pulls reports from database into a file

Interaction: Controller to Database

Vulnerability Level: V2

Risk: Data leakage or incorrect report generation

Mitigation: Use secure database query and restrict data access

Step 4: System provides access to report file

Interaction: System to Administrator UI

PennStateSoft	Version: 1.0
	Date: 06/26/26

Vulnerability Level: V2

Risk: Unauthorized file access or sensitive data exposure

Mitigation: Restrict file access to authenticated administrators only

Step 2A1: Administrator selects Generate Report twice

Interaction: Admin to UI

Vulnerability Level: V3

Risk: Duplicate report generation request

Mitigation: Prevent duplicate submissions and check existing report status

Step 2A2: System confirms report already generated

Interaction: System to Admin UI

Vulnerability Level: V1

Risk: No security risk

Mitigation: Display safe message only

3. Team Members Log Sheets

3.1 Victoria Worthington

date	task	duration
06/27/26	Group meeting	1h 00m
07/01/26	Section 2.2, 2.3	1h 25m
07/02/26	Section 2.3	0h 10m
07/02/26	Usecase-02, Group Meeting	0h 22m
07/03/26	Usecase-09, 15, 17, 19	0h 30m
07/04/26	Usecase diagrams 02, 09, 15, 17, 19	1h 45m
07/05/26	Sequence diagrams 02, 09, 15, 17, 19	2h 15 m
07/06/26	Usecase Fail Case 02, 09, 15, 17, 19	0h 10m
Total :		7h 55m

3.2 Samuel Shumake

date	task	duration
06/26/2026	Sections 1.1, 1.2, and 1.3	0h 45m
06/27/2026	Group meeting, use case findings	1h 00m
07/01/2026	Section 2.5, Usecase-01	0h 45m
07/02/2026	Usecase-06, Usecase-15, Usecase-19. Team meeting	1h 20m
07/05/2026	Usecase and sequence diagrams for Usecase-01,06,14,16,18 and overview use case diagram	2h 00m
07/06/2026	Formatting document	0h 15m
Total :		6h 05m

PennStateSoft	Version: 1.0
	Date: 06/26/26

3.3 Lianjie Sun

date	task	duration
06/26/2026	Section 1, 2.1, 2.7	0h 40m
06/27/2026	Group Meeting	1h 00m
06/30/2026	Section 2.4, 2.7 (use cases 3, 5)	1h 45m
07/01/2026	Section 2.7 (use case 8, 10), 2.8	3h 00m
07/03/2026	Section 2.7 (use case 13)	0h 40m
07/05/2026	Final Review, Document Formatting	2h 00m
07/06/2026	Final Formatting Document	1h 00m
Total :		10h 05m

3.4 Tiffany Hart

date	task	duration
06/27/2026, 07/02/2026	Group Meeting	1h 15m
07/01/2026, 07/02/2026	Section 2.6. 1.5	2h 0m
07/03/2026	Use Case-04, Use Case-07, Use Case-12, Use Case-13,	2h 45m
07/04/2026,07/05/2026	Document Formatting	1h 30m
Total :		7h 30m